

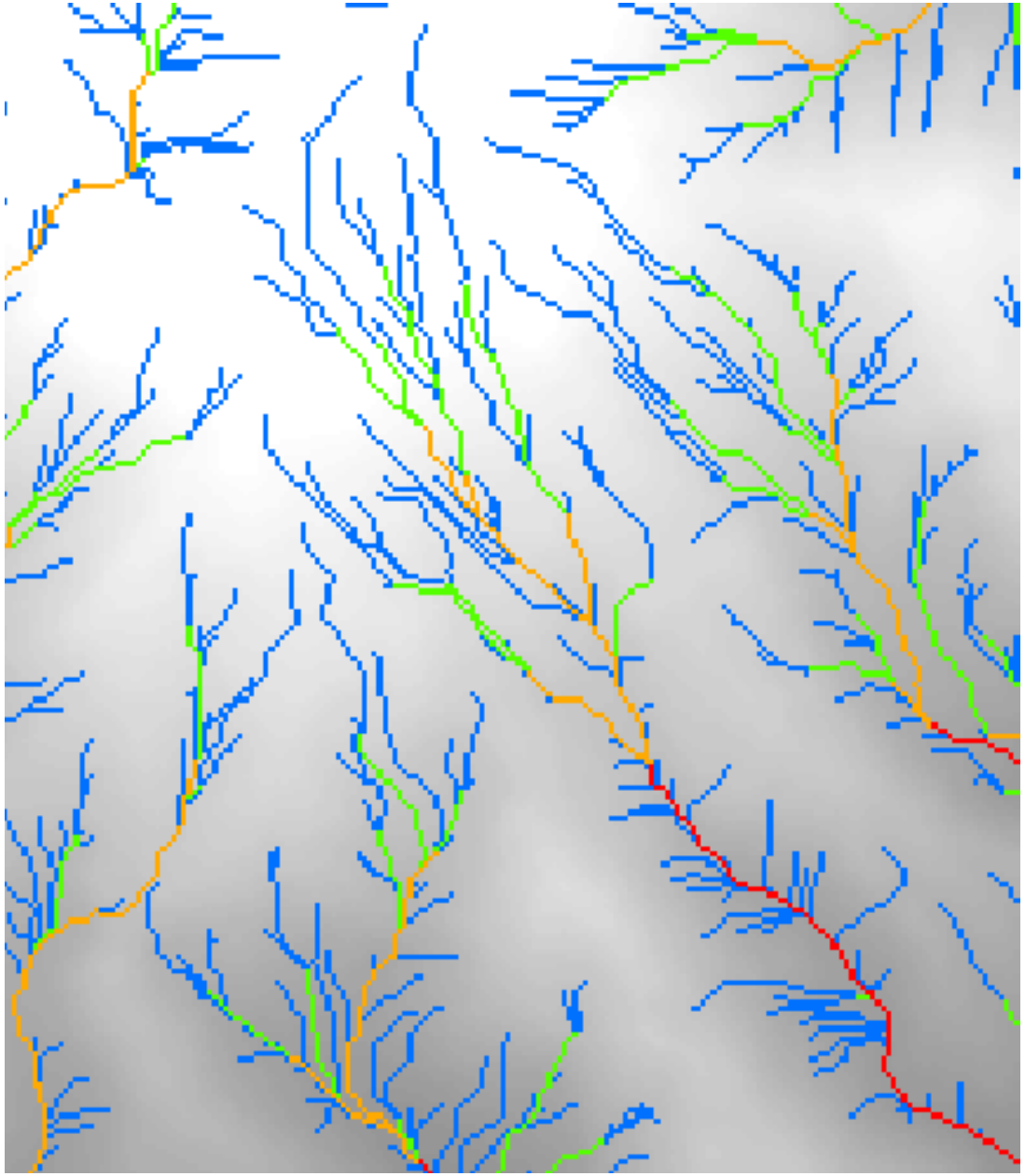
LAHARZ Streams and Cones: the details

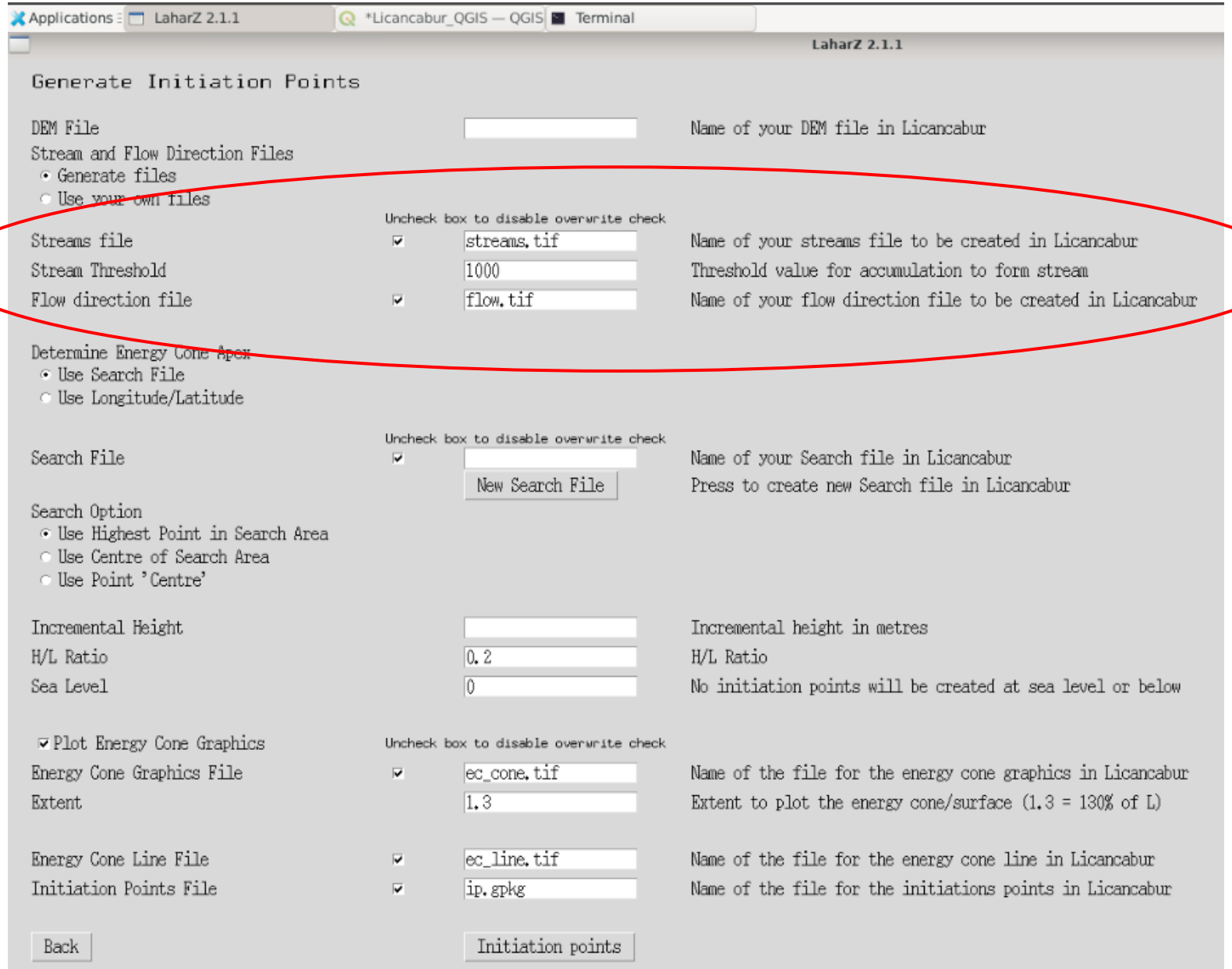
Stream Network

- LAHARZ generates stream network using the DEM, and a user defined stream threshold
- This is the number of upstream cells that have to flow into a cell to consider it a stream cell. A smaller number here will make a stream network with more branches and longer streams, a higher number creates a simpler stream network with fewer branches
- The network should generally match published maps, but should represent the *current* stream network

High stream threshold (6000): shorter stream network with fewer branches

Low stream threshold (50): longer stream network with more branches




Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
Generate Initiation Points
DEM File
Name of your DEM file in Licancabur
Stream and Flow Direction Files
Generate files
Use your own files
Uncheck box to disable overwrite check
Streams file
streams.tif
Name of your streams file to be created in Licancabur
Stream Threshold
1000
Threshold value for accumulation to form stream
Flow direction file
flow.tif
Name of your flow direction file to be created in Licancabur
Determine Energy Cone Apex
Use Search File
Use Longitude/Latitude
Uncheck box to disable overwrite check
Search File
Name of your Search file in Licancabur
New Search File
Press to create new Search file in Licancabur
Search Option
Use Highest Point in Search Area
Use Centre of Search Area
Use Point 'Centre'
Incremental Height
Incremental height in metres
H/L Ratio
0.2
H/L Ratio
Sea Level
0
No initiation points will be created at sea level or below
Plot Energy Cone Graphics
Uncheck box to disable overwrite check
Energy Cone Graphics File
ec_cone.tif
Name of the file for the energy cone graphics in Licancabur
Extent
1.3
Extent to plot the energy cone/surface (1.3 = 130% of L)
Energy Cone Line File
ec_line.tif
Name of the file for the energy cone line in Licancabur
Initiation Points File
ip.gpkg
Name of the file for the initiations points in Licancabur
Back
Initiation points

VDAP

USGS



Elevation (DEM)

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78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

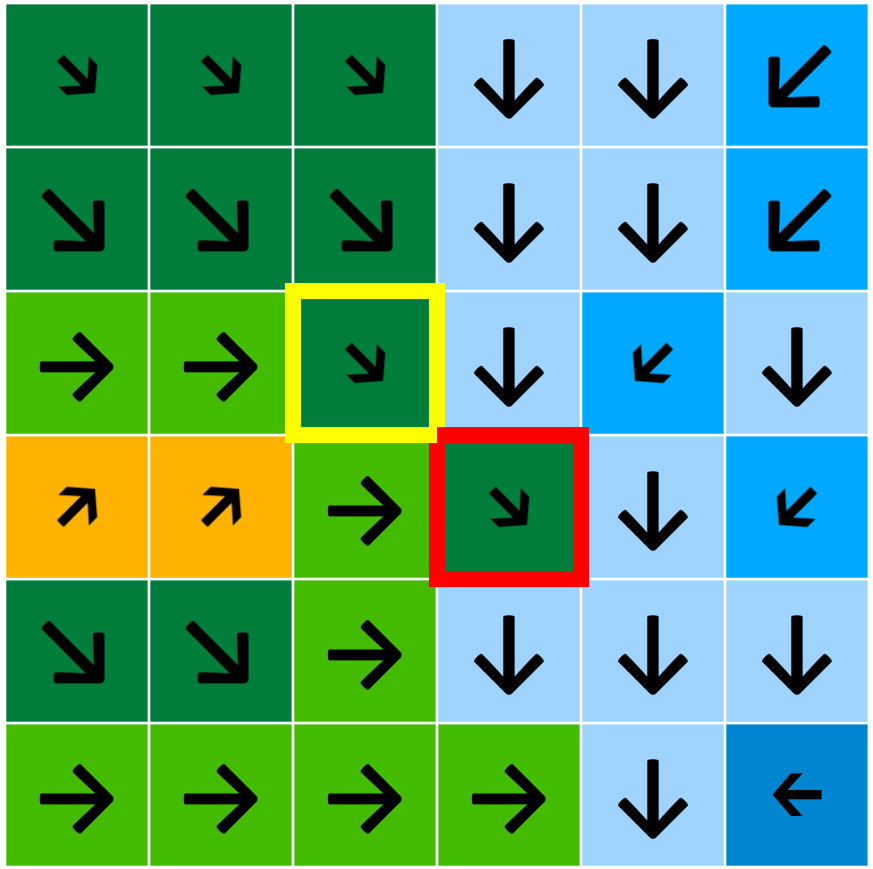
Elevation
(DEM)

Elevation→Flow direction

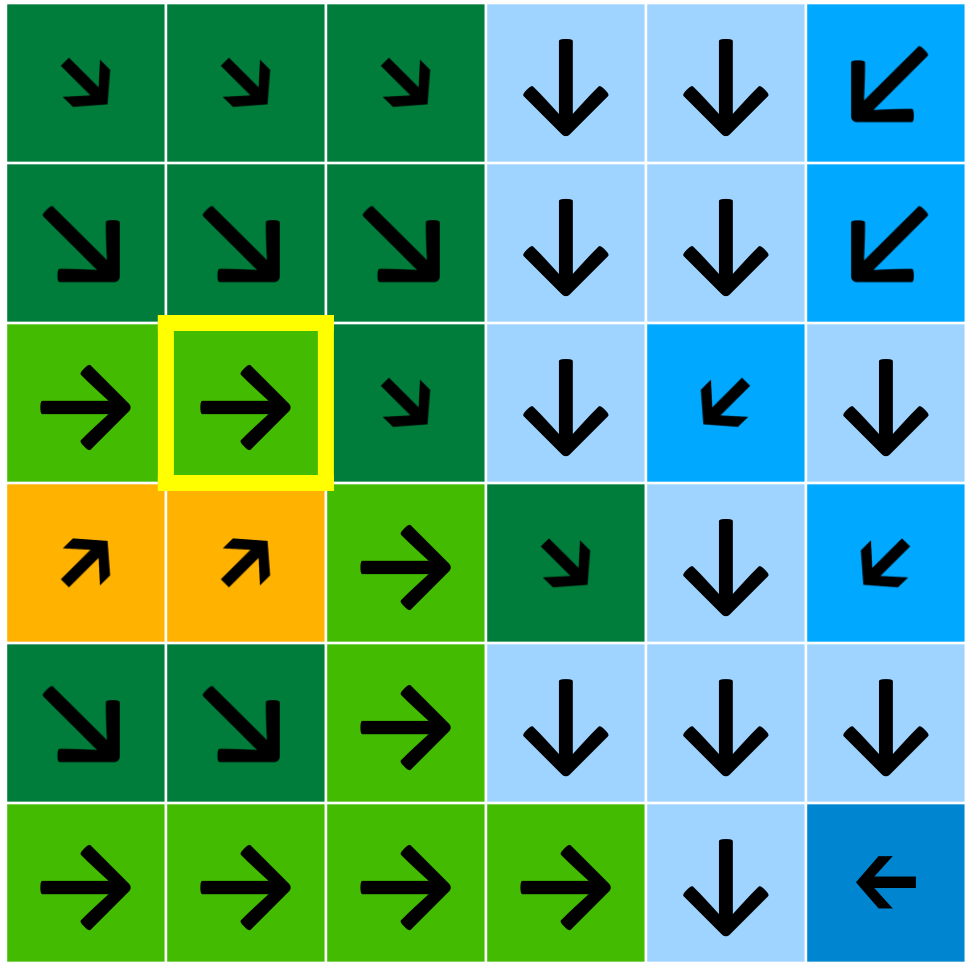
Where would a drop of water that fell on the yellow square go?



Elevation (DEM)



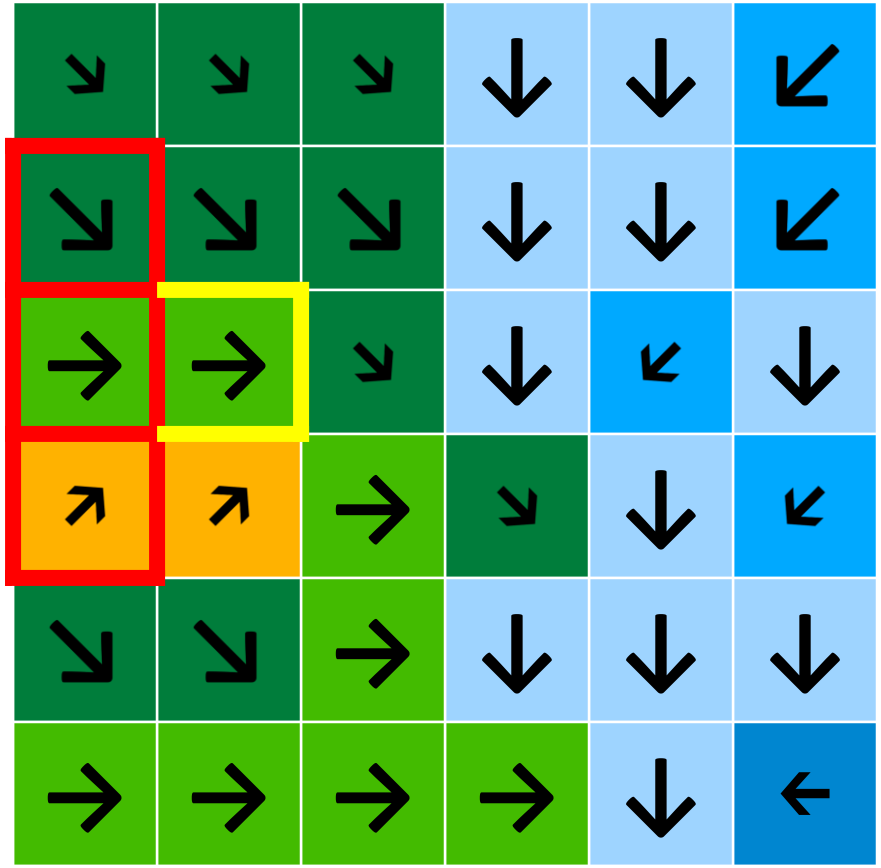
Flow direction



Flow direction

Flow direction → Flow accumulation

How many cells flow into the yellow square?



Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

Flow accumulation → Stream network

With a stream threshold of 2 what does the stream network look like?

Stream threshold = 2

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0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 2

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

Stream threshold = 6

Flow accumulation → Stream network

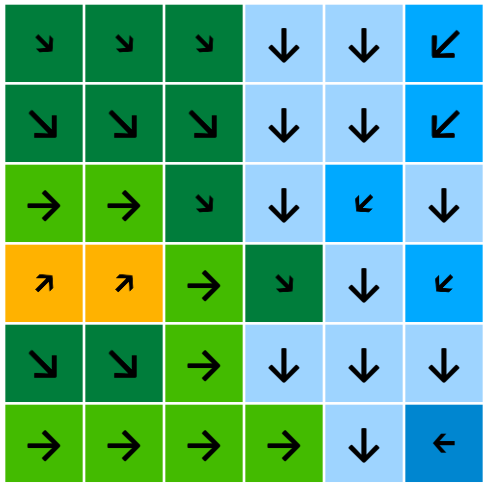
With a stream threshold of 6 what does the stream network look like?

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 6

78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

Elevation (DEM)



Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 2

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 6

Lower stream threshold: longer stream network with more branches

Higher stream threshold: shorter stream network with fewer branches

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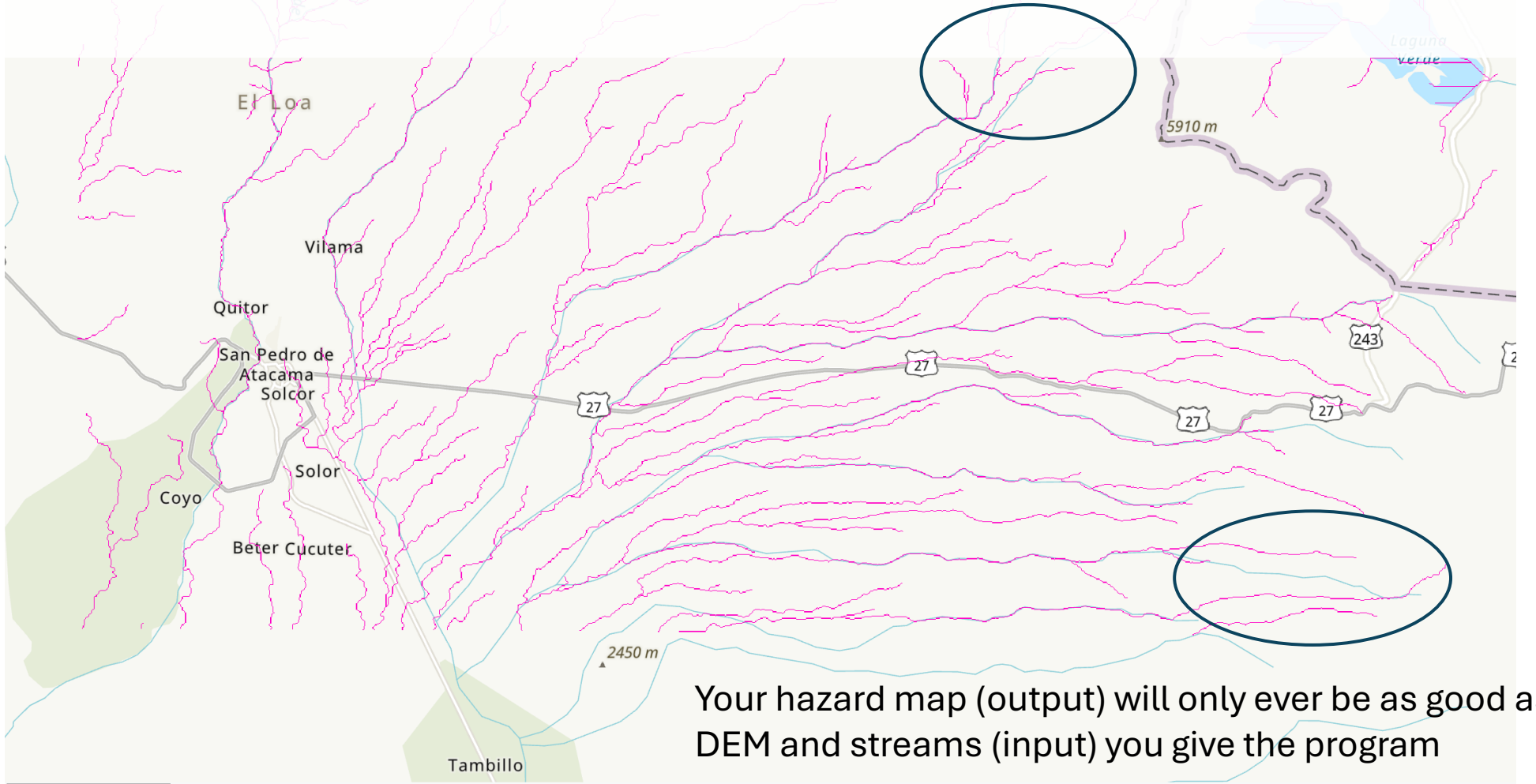
How to decide on the stream threshold

- Compare your stream network to a recent topographic map
- Compare your stream network to a recent satellite image
- Compare your stream network to field measurements
- Examine the stream network in locations where lahars/flash floods have been known to spill over into other channels
- If you are happy with you ΔH/L cone, but the streams don't intersect, you can try lowering the stream threshold, but this might also be an indication to think about your ΔH/L value!

Evaluate the computer-generated streams to a basemap or (better) an authoritative map made locally and recently.

Reasons why there may be differences:

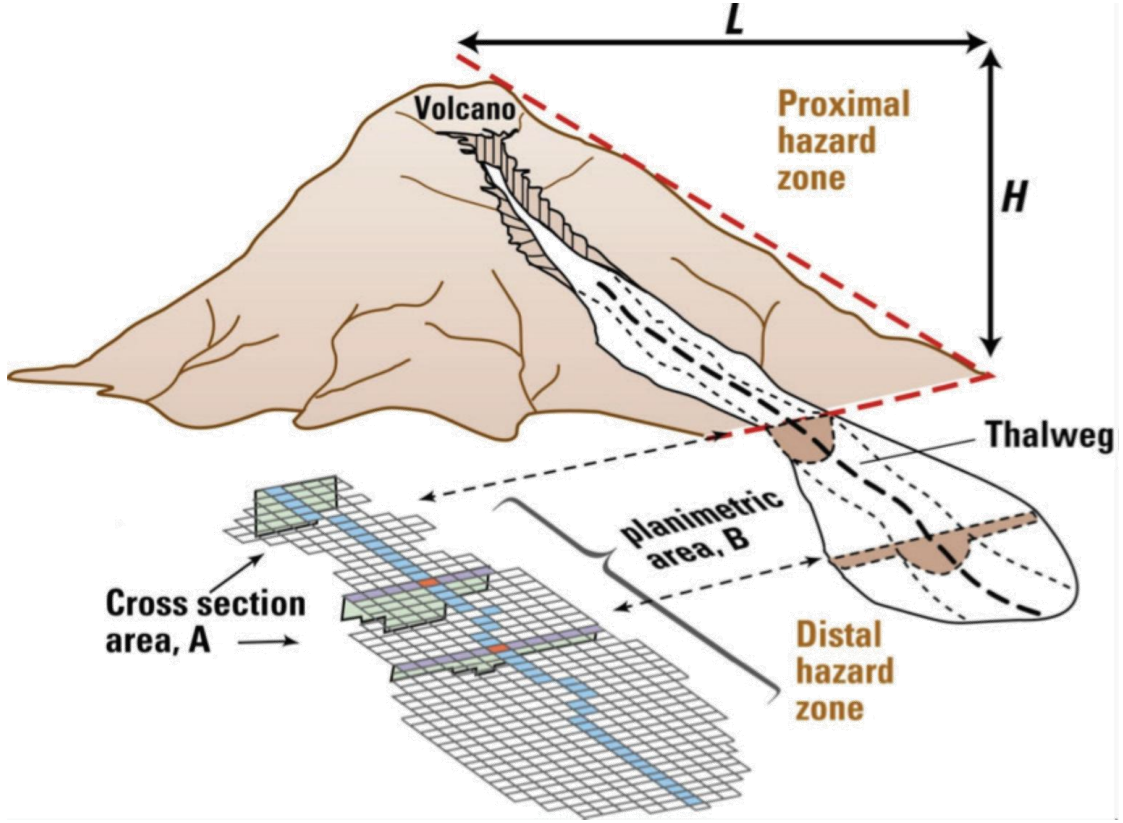
- Landscapes change over time, seasonally or catastrophically
- Man-made features like paving, canals, dams or bridges may alter the topography
- Map made at the wrong scale to be accurate at this local- regional scale
- DEM may be older and may not reflect changes
- Errors



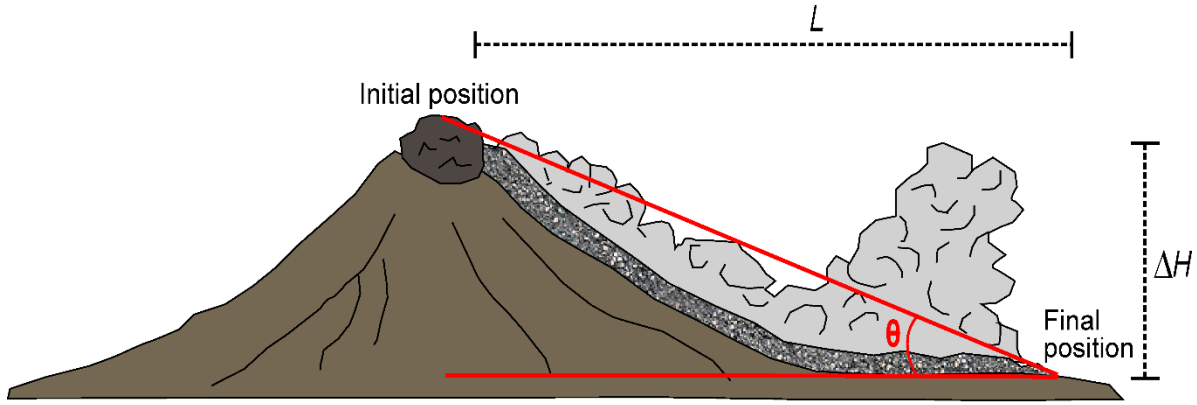
Your hazard map (output) will only ever be as good as the DEM and streams (input) you give the program

ΔH/L ratio
LAHARZ proximal hazard zone vs. energy cone for PDCs

For LAHARZ



For PDCs

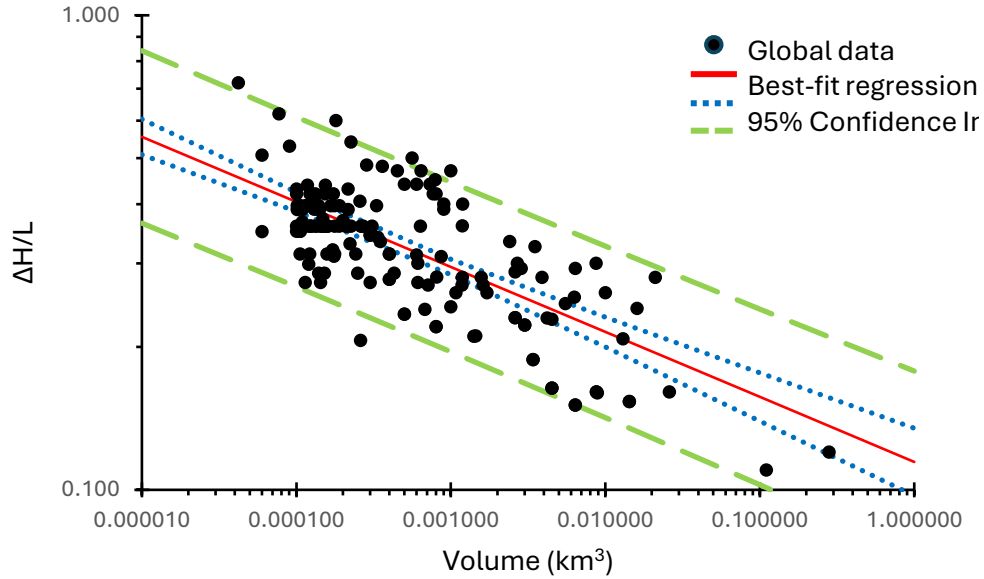


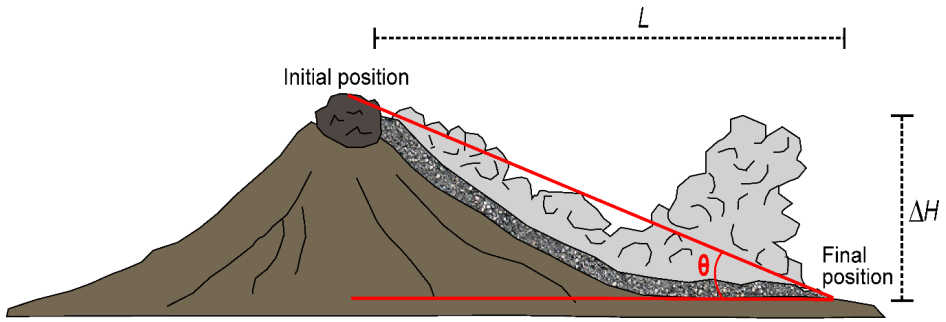
H/L = energy cone, coefficient of friction, Heim coefficient, H/L ratio

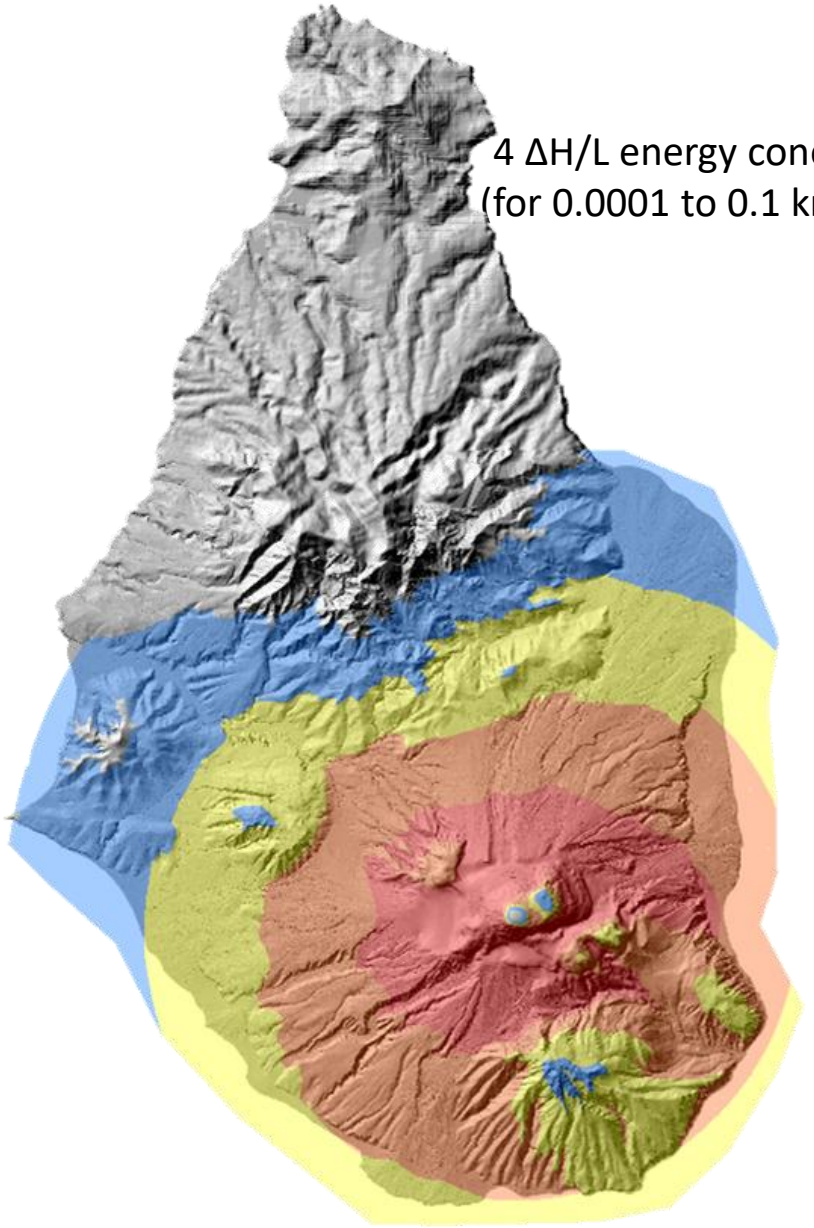
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USGS

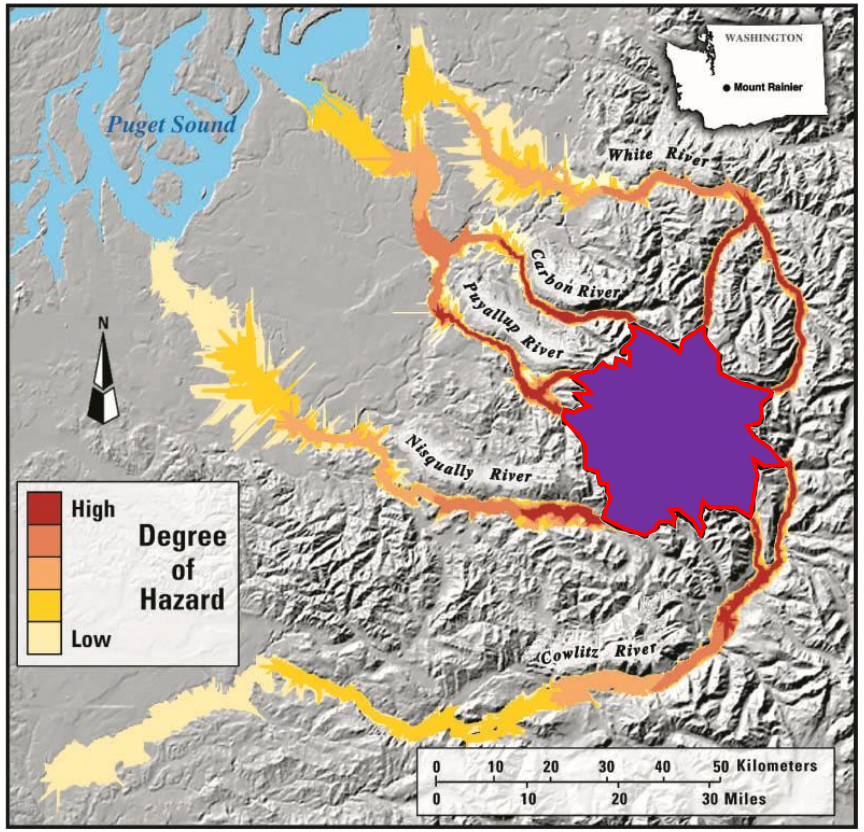
Relationship between PDC volume and H/L is frequently used in energy cone modeling of PDCs
(we'll talk about this more later!)







4 ΔH/L energy cones (for 0.0001 to 0.1 km^3)





On a hazard map, the Proximal Hazard Zone, or highest hazard zone, is the most hazardous, most destructive area around a volcano because of steep slopes and proximity to the summit.

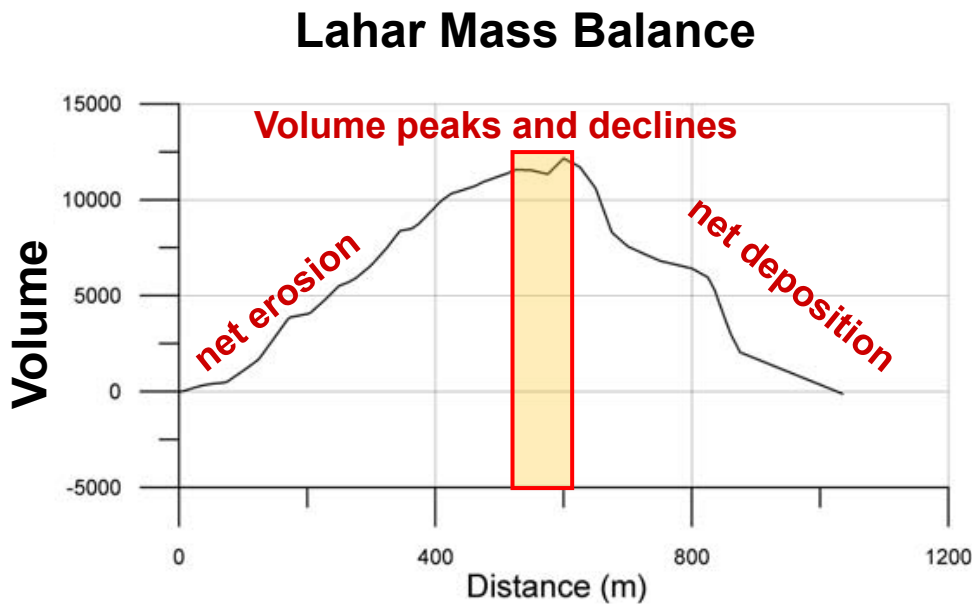
Proximal hazard zones usually include smaller pyroclastic flows, volcanic bombs, vent lava flows and tephra fall.

For LAHARZ, creating a proximal hazard zone is also a convenient way to determine start locations for modeling lahars.

H/L Cone for LAHARZ

Why use H/L Cone for lahars?

- Lahars may initiate anywhere on the volcano edifice. We can't know an exact start location in advance.
- Loose pyroclastic flow deposits and ashfall are often the source material for lahars and are most abundant within the H/L cone.
- Lahars gain volume (entrain material) on steep slopes.



- So, we start modeling lahars by estimating where on the flanks the flow volume is greatest.
- We do this by defining an H/L cone that marks the transition between erosion and deposition. That is, inside the H/L cone, the volume is still increasing. Outside the H/L cone, the volume is maintained or decreasing (depositing).
- How do we identify an H/L cone that represents this transition?
    - from net erosion to net deposition
    - from steep slopes to moderate slopes (20°-30°), that is, find the break in slope
    - from pyroclastic debris fans to more channelized flows

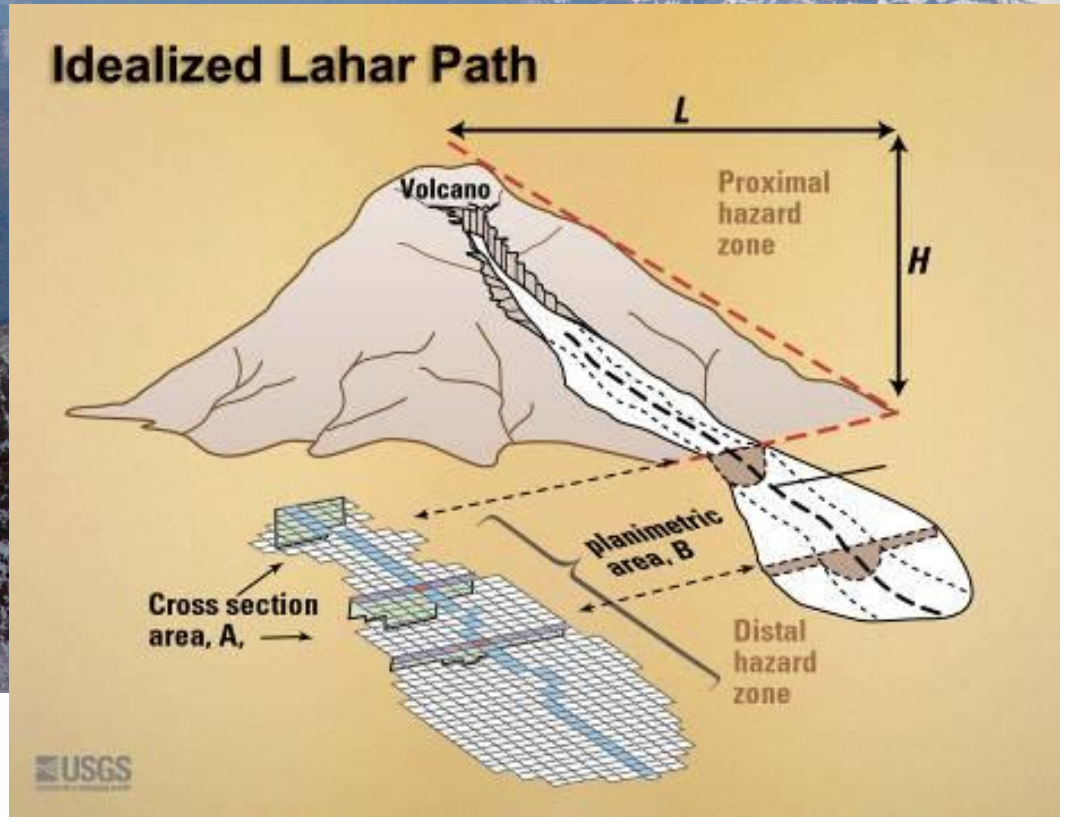
Mt. Rainier, west side

L

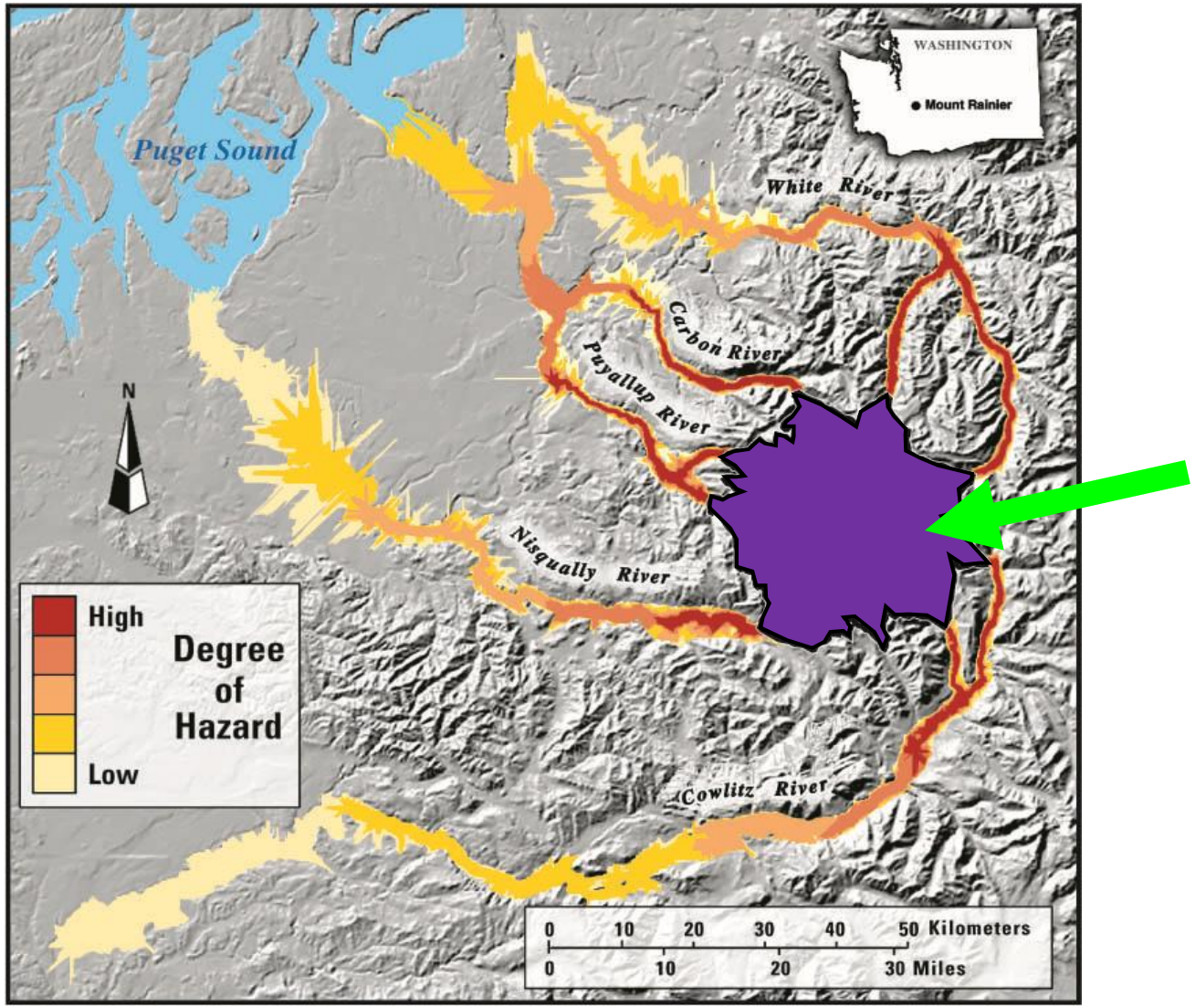
V

H

Begin flow hazard-zone computation where deposition first exceeds erosion

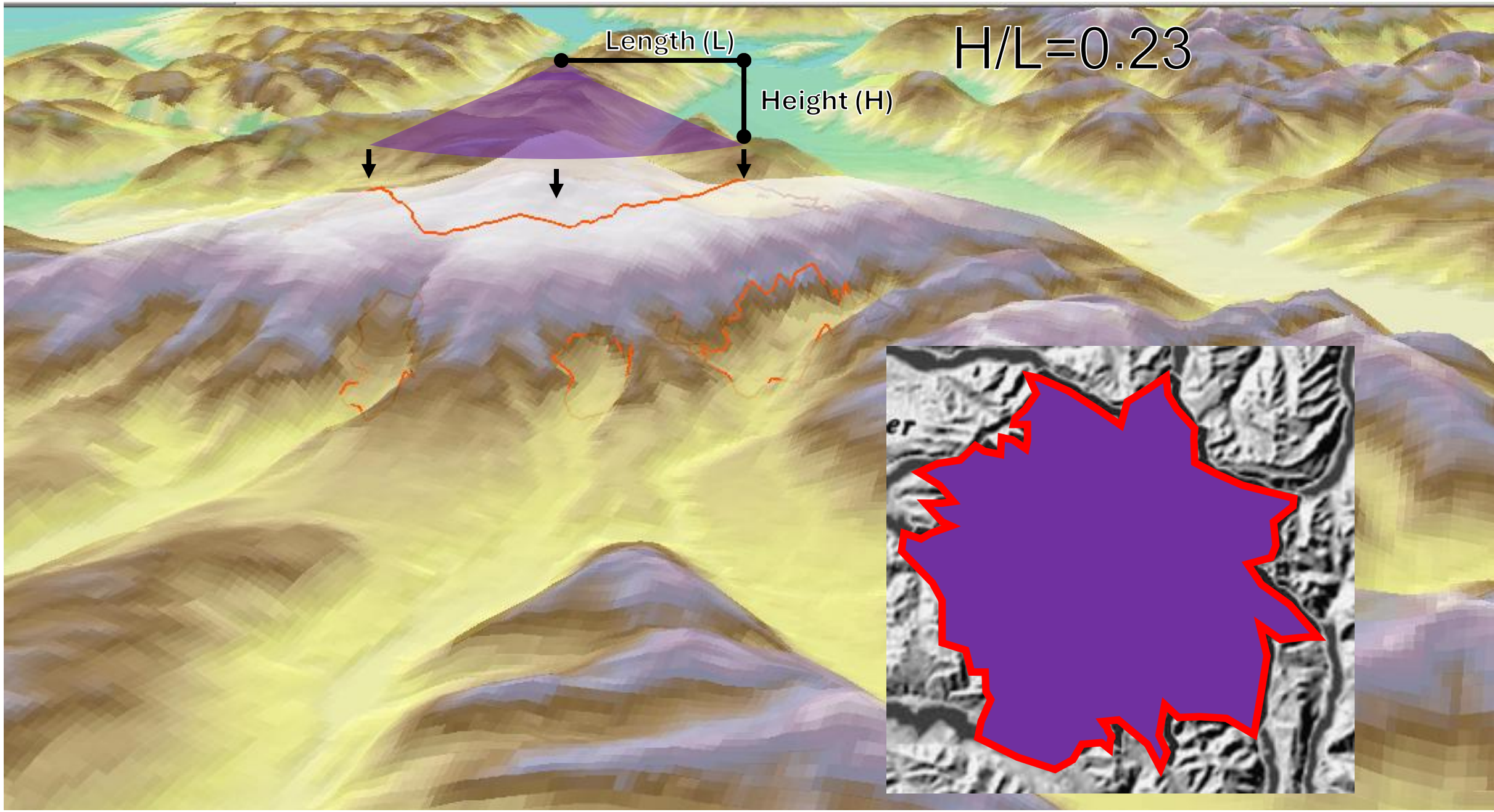


Lahar hazard zones computed for Mt. Rainier with proximal hazard zone defined by *H/L* = 0.23



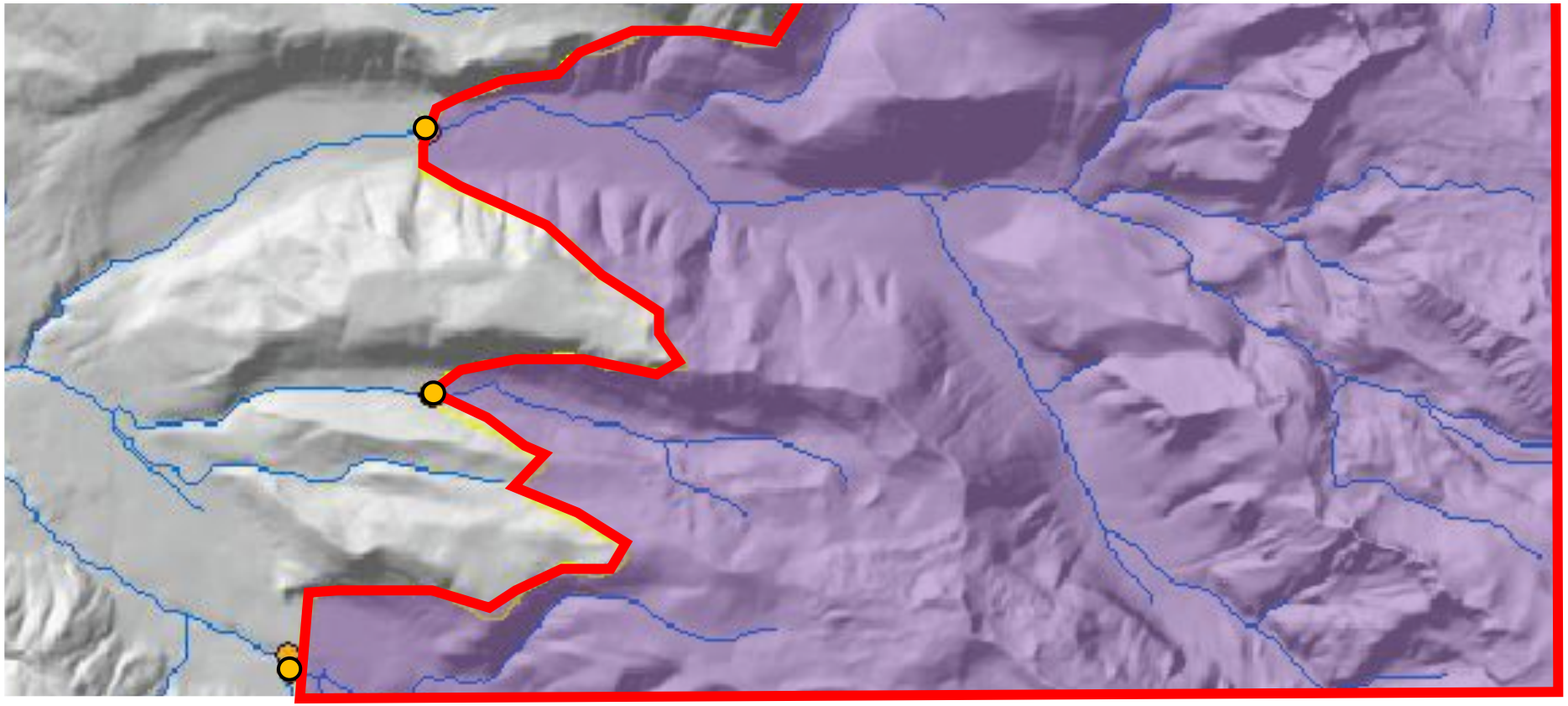
VDAP

USGS



VDAP

USGS



VDAP

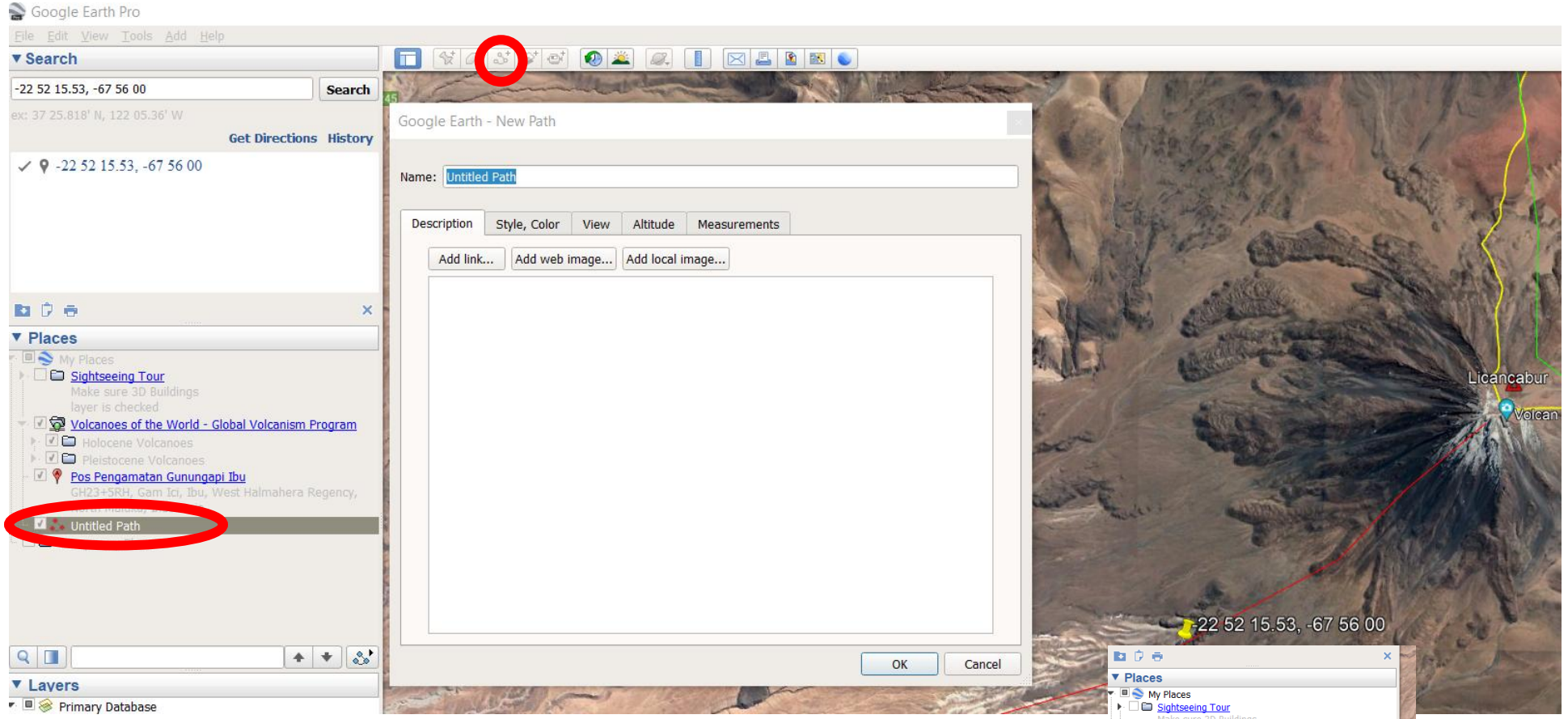
USGS

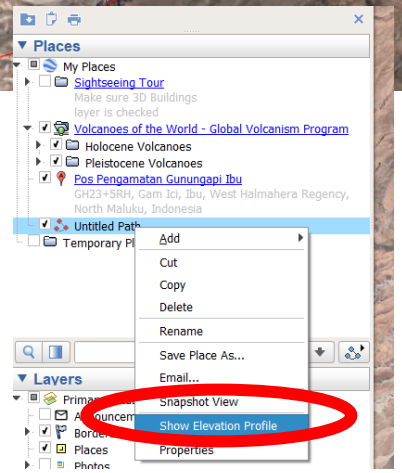
Okay, so how do we choose the H/L value for LAHARZ?

Changes in slope and the extent of PDC deposit fans (where most smaller PDC deposits are) are useful for choosing an H/L ratio

- **Visualize volcano in 3-D:**
  - Find photos (Google image search, GVP)
  - Use Google Earth to get oblique views, fly around, and make a topographic profile.
  - Look for change in slope
- **Look at the geomorphology:**
  - landforms, farming, land use
- **Look for PDC units in the field**
- **Examine satellite imagery**
- **Use global record of analog volcanoes (PDC H/L values)**
- **Start with H/L of 0.3 and refine**

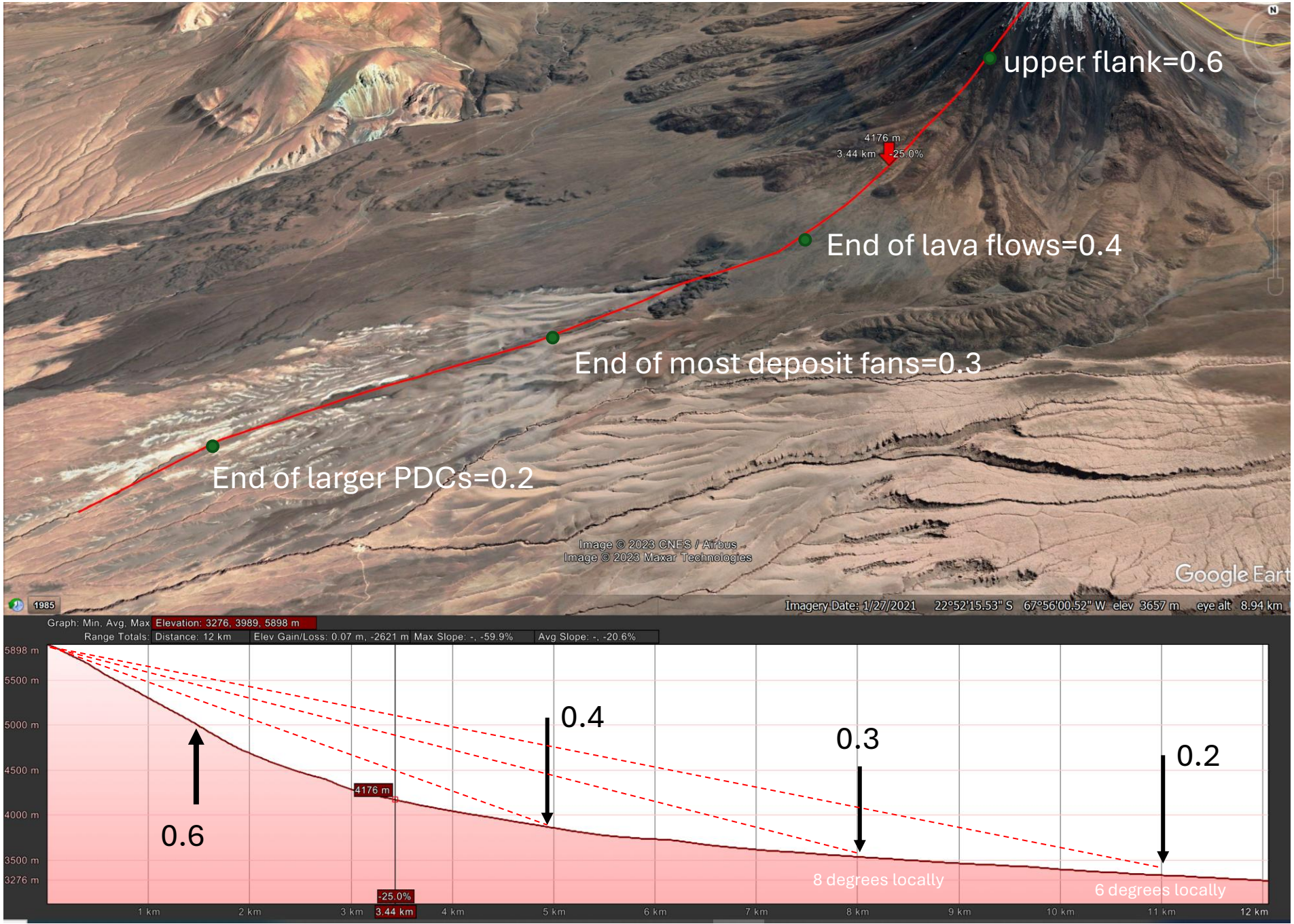
Visualize volcano in 3-D



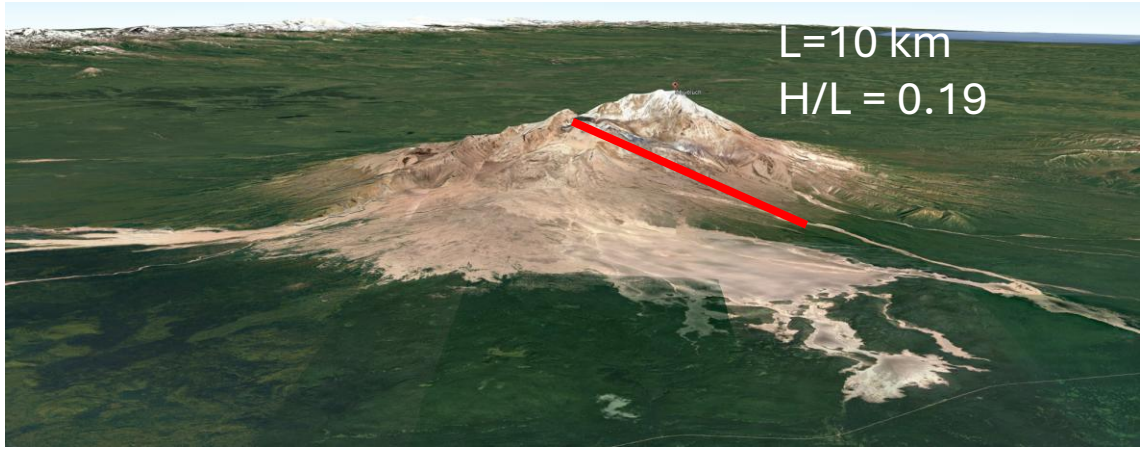


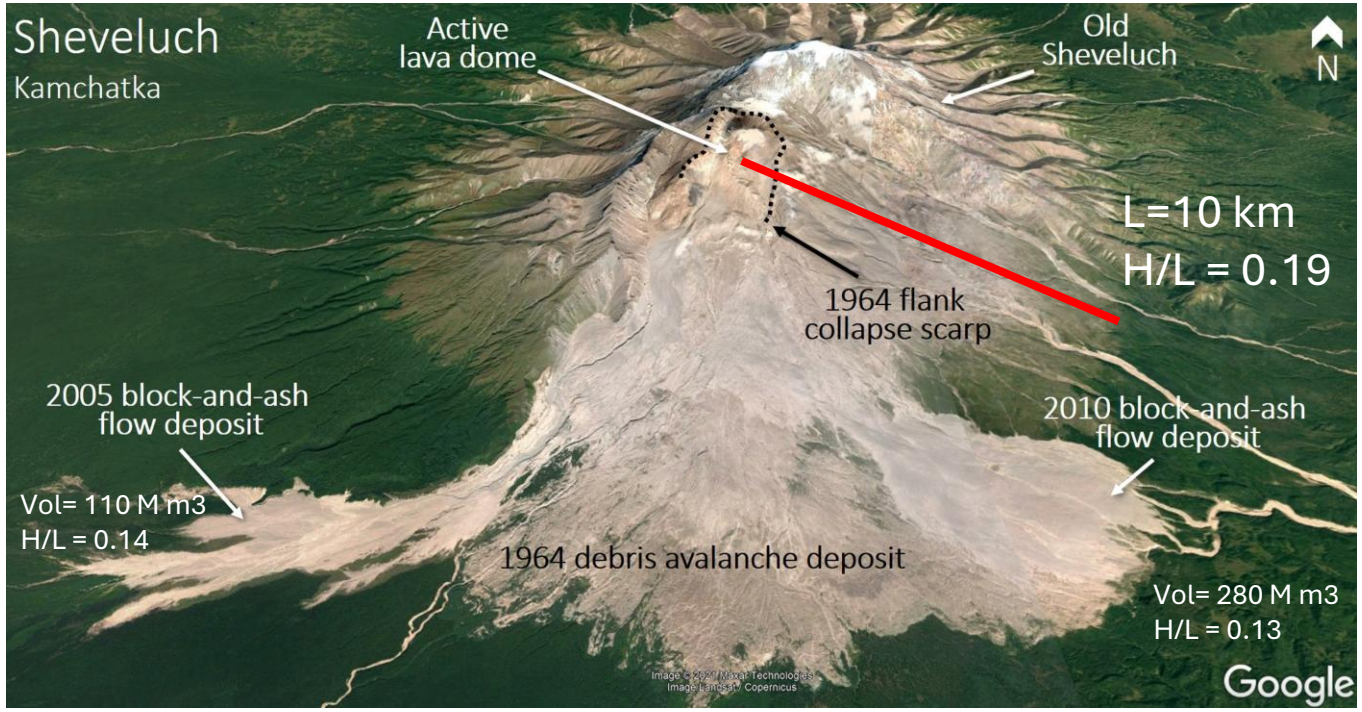
- Click on the “Add path” button
- Click on the map to make a path
- Name and save the path
- Right click on the path name under “Places”
- Choose “show elevation profile”
- Be careful: the slope values given in Google Earth are local, not H/L from the summit

Visualize volcano in 3-D



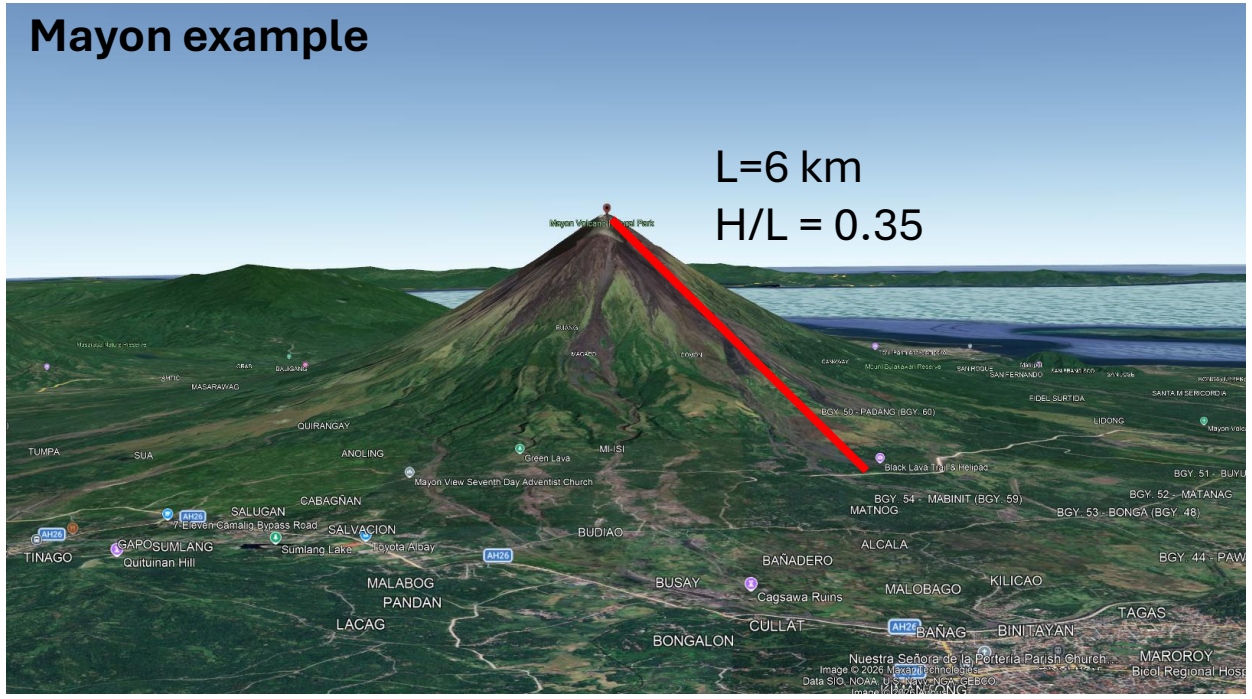
Sheveluch example

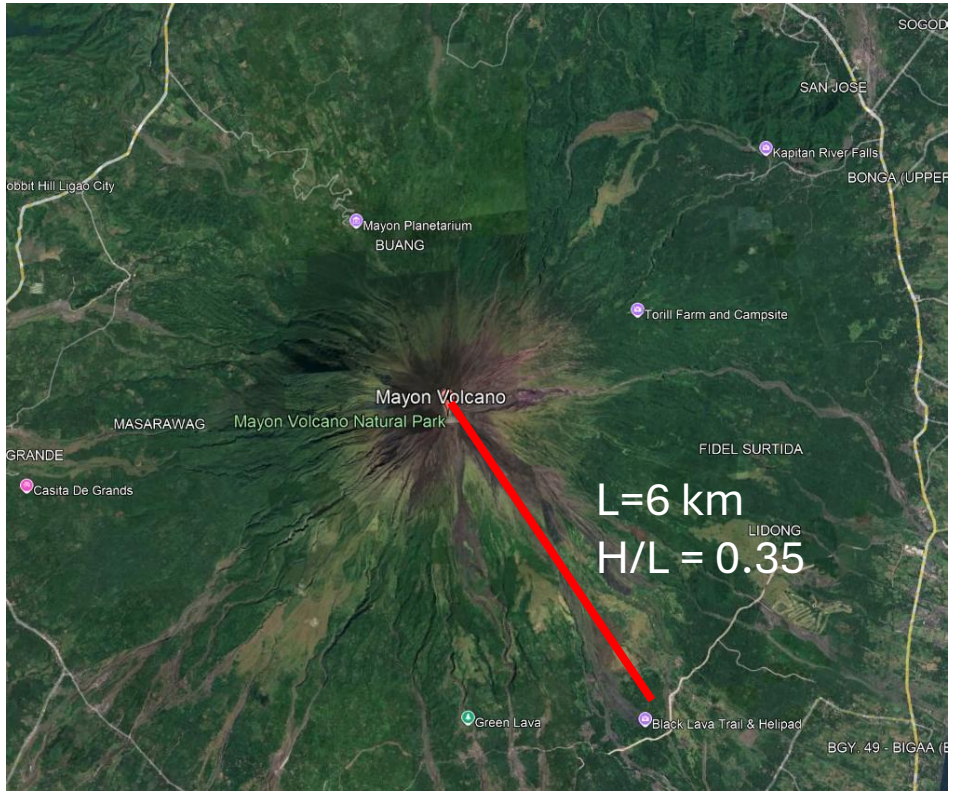






- **Look at the slope**
- **Look at majority of PDC deposits**
- **Look for landcover changes**



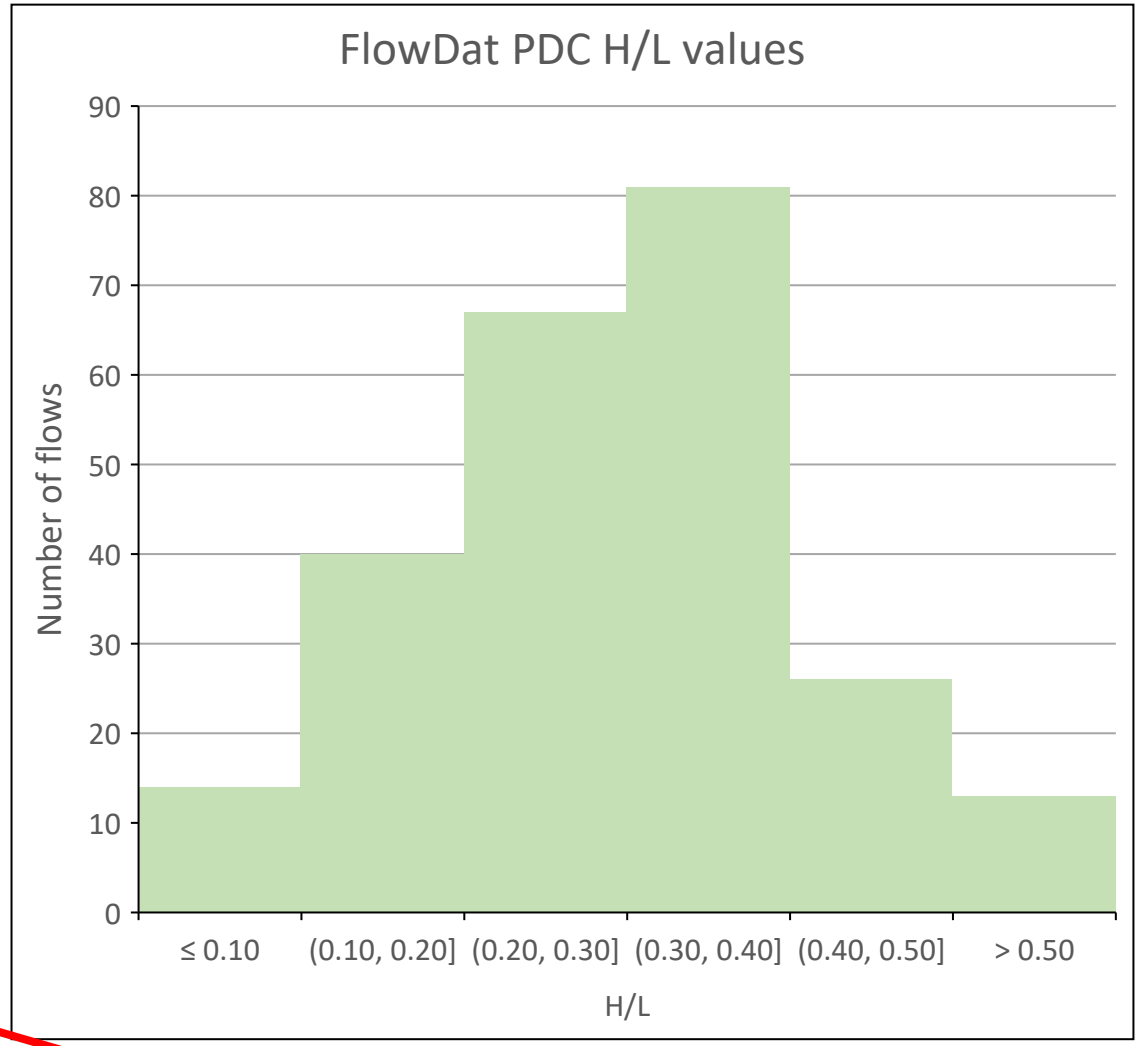




- **Look at the slope**
- **Look at majority of smaller PDC deposits**
- **Look for landcover changes**

Use global record of analog volcanoes (PDC H/L values)

- ΔH/L values range from ~0.04 for very large ignimbrites to 0.8 for very small dome-collapses or rockfalls
- Mean: 0.30
- Median: 0.30
- Most (80%) of ΔH/L values fall between 0.15 and 0.44



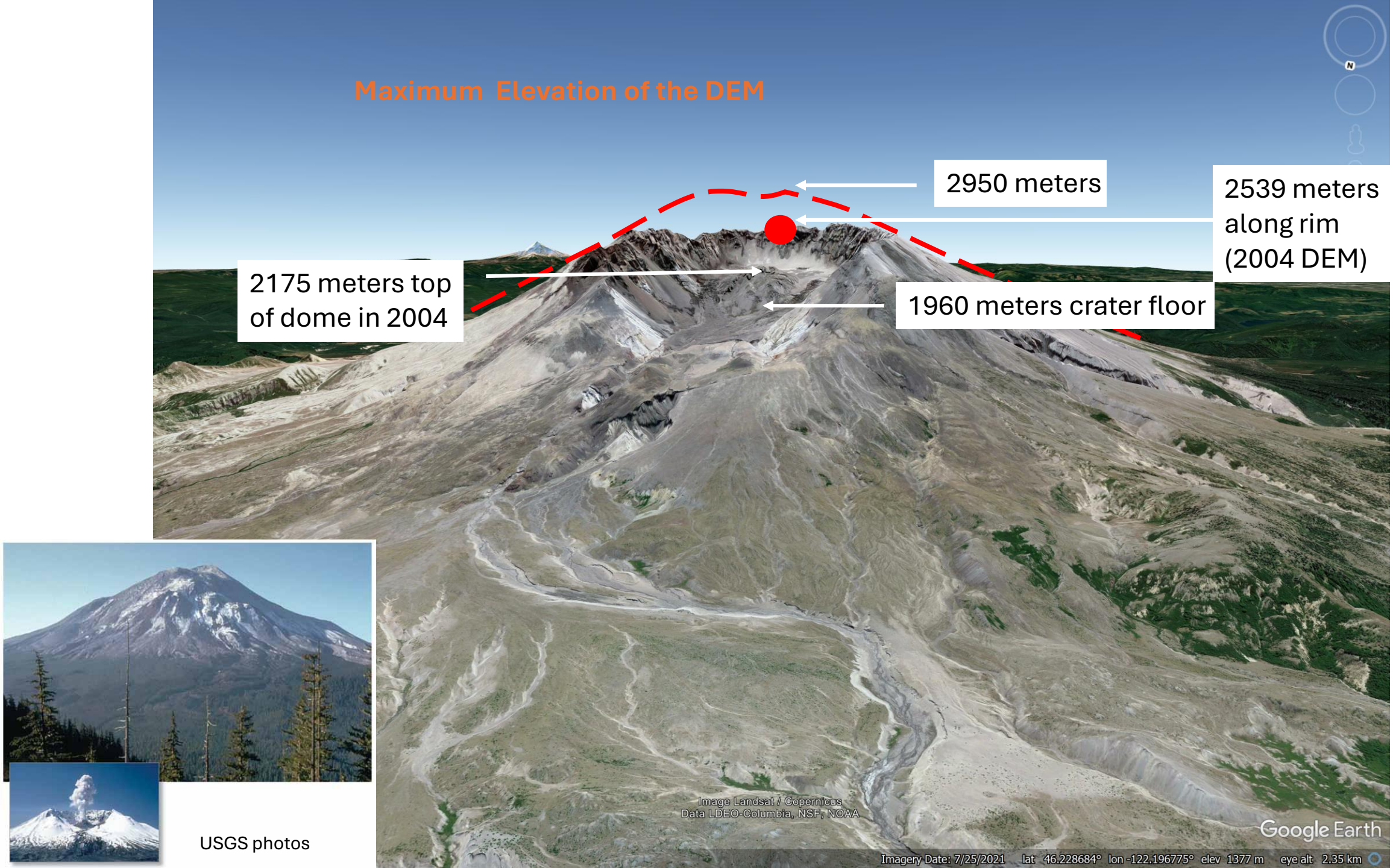
Start with H/L of 0.3 and refine

Special Cases: not a nice pointy volcano?

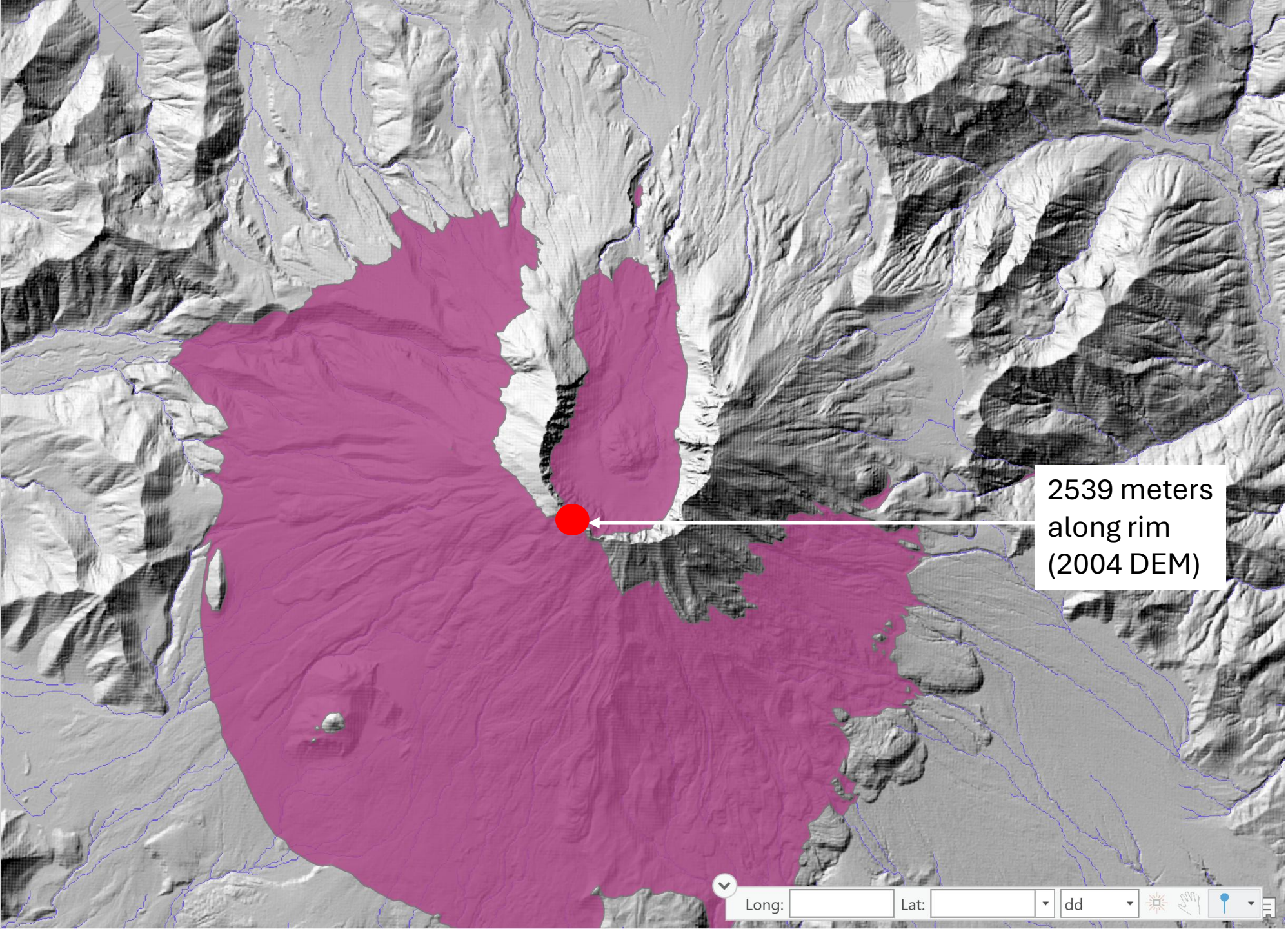


Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ




Maximum Elevation of the DEM
2950 meters
2539 meters along rim (2004 DEM)
2175 meters top of dome in 2004
1960 meters crater floor
Image Landsat / Copernicus
Data LDEO-Columbia, NSF, NOAA
Google Earth
Imagery Date: 7/25/2021 lat 46.228684° lon -122.196775° elev 1377 m eye alt 2.35 km

USGS photos



This ΔH/L cone uses the highest point on the DEM, a point on the crater rim. This ΔH/L cone has holes and looks weird!

To-restablish the summit:
2950 m – 2175 m = add ~800 meters

2950 meters

2539 meters along rim (2004 DEM)

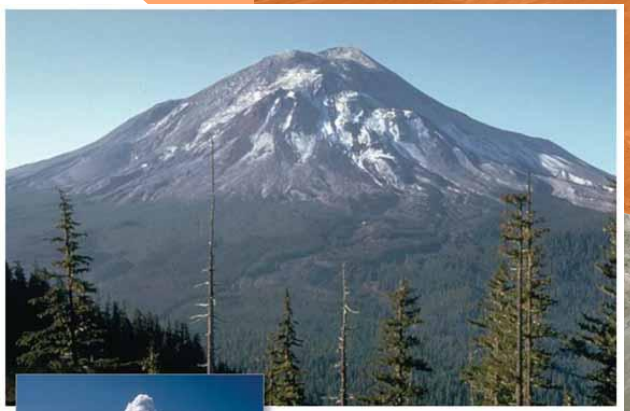
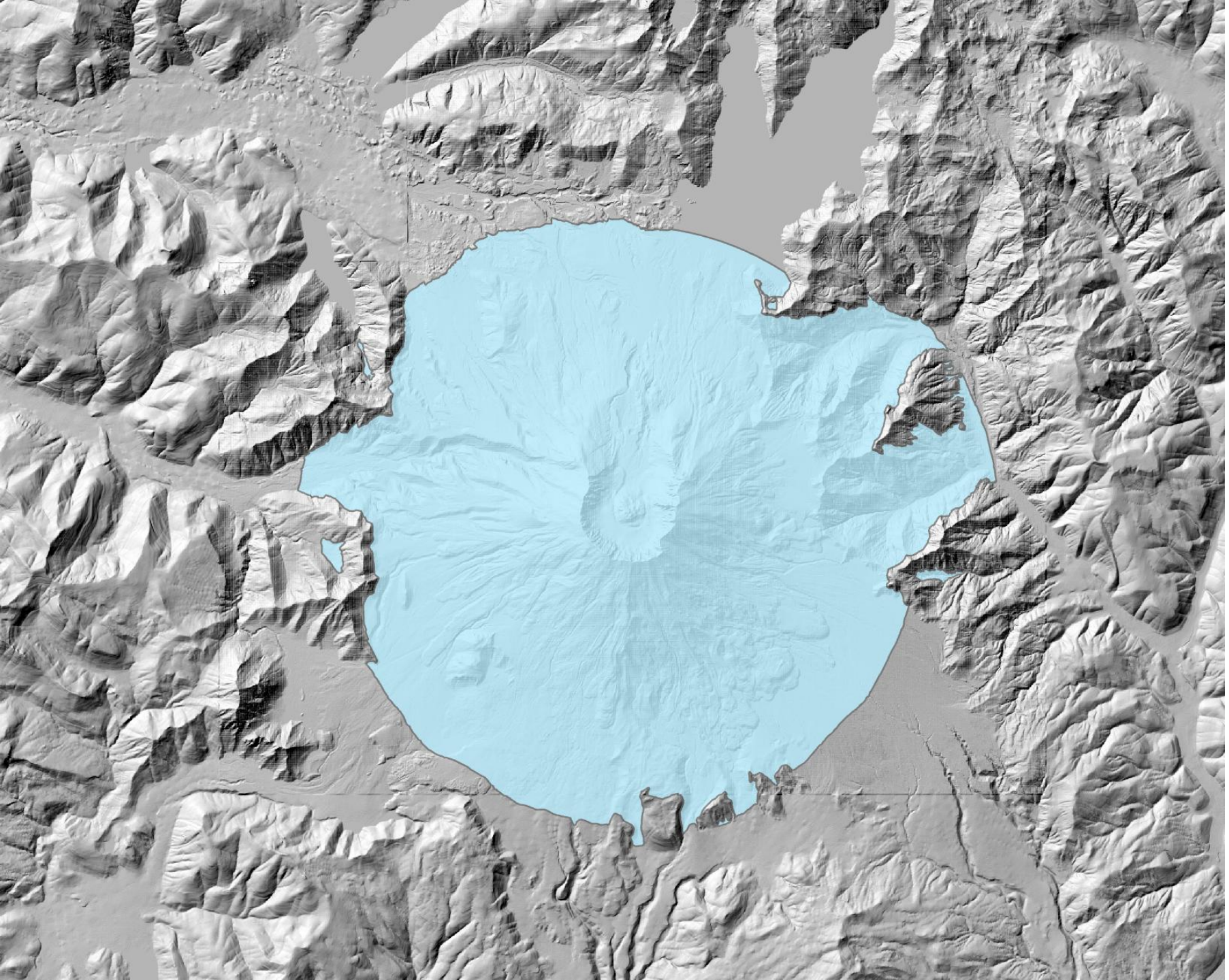
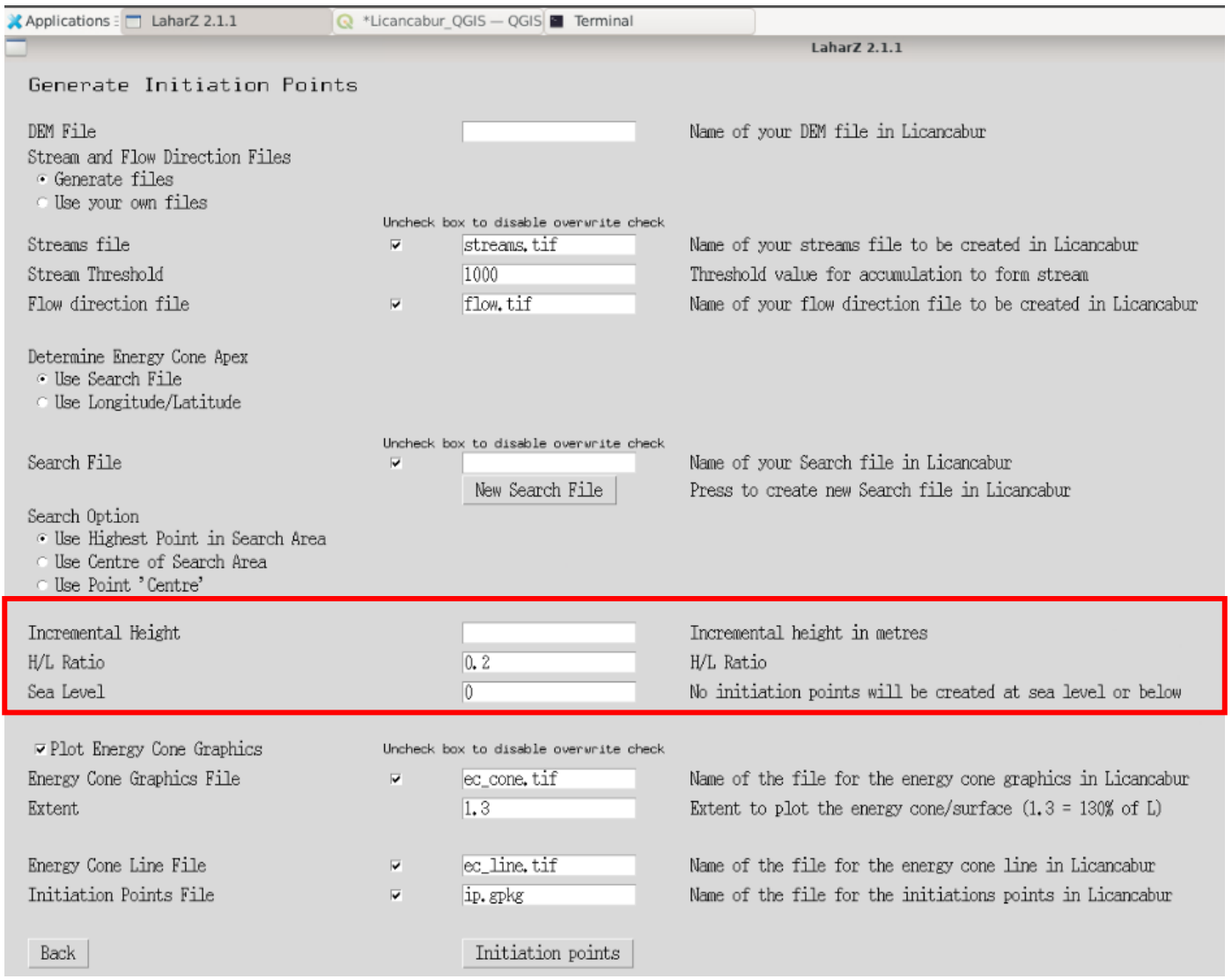


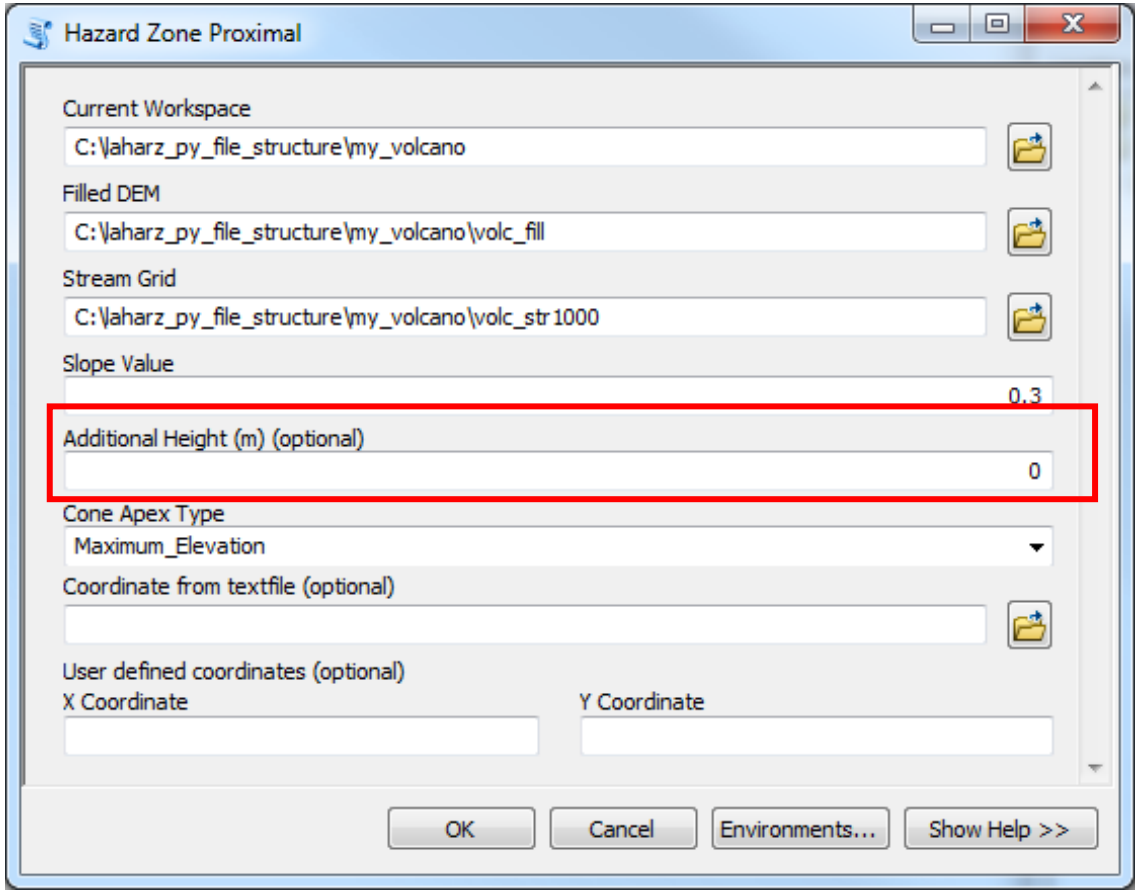


Image Landsat / Copernicus
Data LDEO-Columbia, NSF, NOAA
Google Earth
Imagery Date: 7/25/2021 lat 46.228684° lon -122.196775° elev 1377 m eye alt 2.35 km



This ΔH/L cone adds ~800 m above the center of the crater. This ΔH/L cone has looks pretty good!

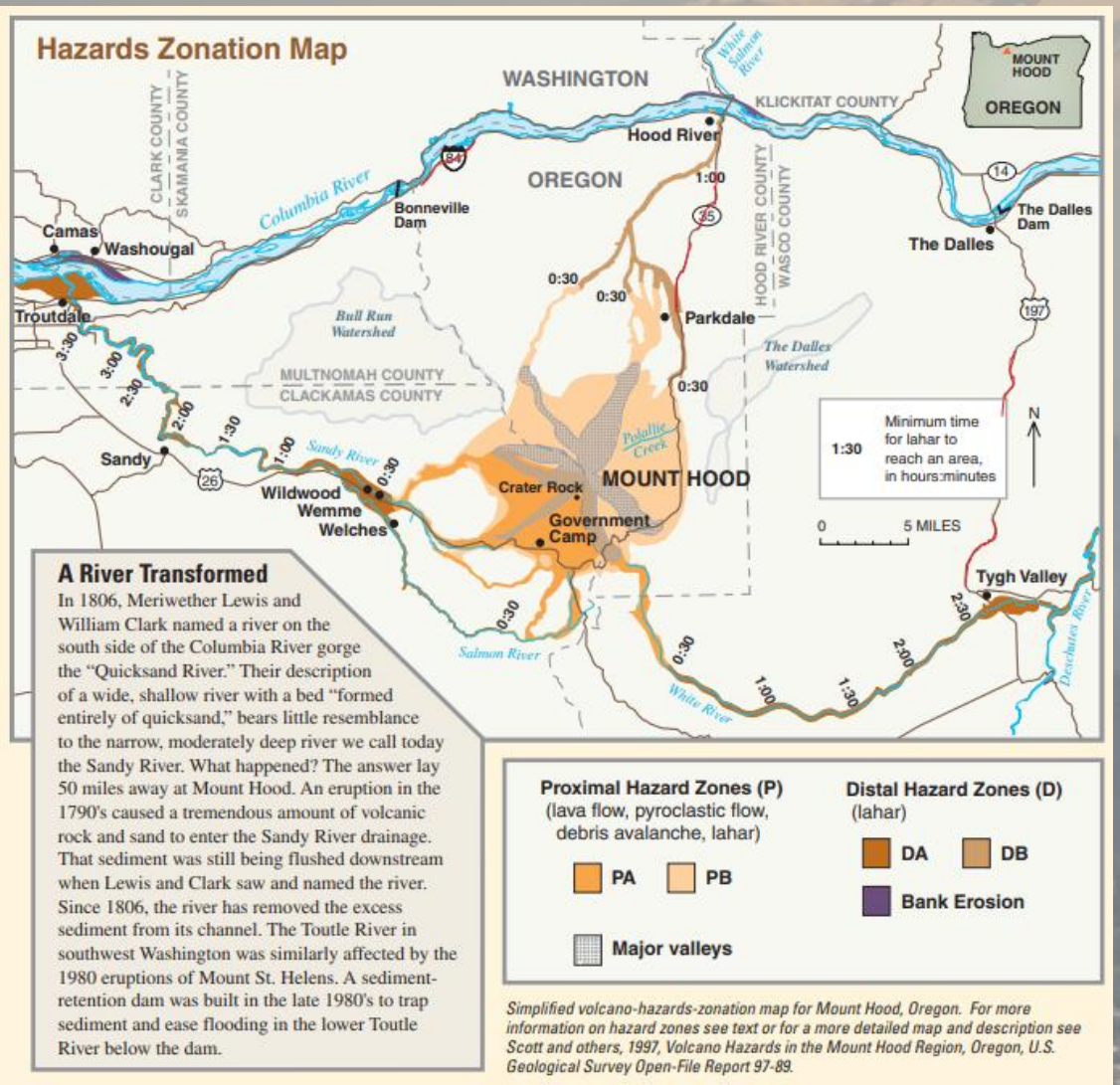

Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
Generate Initiation Points
DEM File
Name of your DEM file in Licancabur
Stream and Flow Direction Files
Generate files
Use your own files
Uncheck box to disable overwrite check
Streams file
streams.tif
Name of your streams file to be created in Licancabur
Stream Threshold
1000
Threshold value for accumulation to form stream
Flow direction file
flow.tif
Name of your flow direction file to be created in Licancabur
Determine Energy Cone Apex
Use Search File
Use Longitude/Latitude
Uncheck box to disable overwrite check
Search File
Name of your Search file in Licancabur
New Search File
Press to create new Search file in Licancabur
Search Option
Use Highest Point in Search Area
Use Centre of Search Area
Use Point 'Centre'
Incremental Height
Incremental height in metres
H/L Ratio
0.2
H/L Ratio
Sea Level
0
No initiation points will be created at sea level or below
Plot Energy Cone Graphics
Uncheck box to disable overwrite check
Energy Cone Graphics File
ec_cone.tif
Name of the file for the energy cone graphics in Licancabur
Extent
1.3
Extent to plot the energy cone/surface (1.3 = 130% of L)
Energy Cone Line File
ec_line.tif
Name of the file for the energy cone line in Licancabur
Initiation Points File
ip.gpkg
Name of the file for the initiations points in Licancabur
Back
Initiation points


Hazard Zone Proximal
Current Workspace
C:\laharz_py_file_structure\my_volcano
Filled DEM
C:\laharz_py_file_structure\my_volcano\volc_fill
Stream Grid
C:\laharz_py_file_structure\my_volcano\volc_str1000
Slope Value
0.3
Additional Height (m) (optional)
0
Cone Apex Type
Maximum_Elevation
Coordinate from textfile (optional)
User defined coordinates (optional)
X Coordinate
Y Coordinate
OK
Cancel
Environments...
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The “proximal hazard zone” on the map

- Conveniently, you can add this energy cone to your map as a “proximal hazard zone” that encompasses gas hazards, ballistics, and small PDCs
- For volcanoes that produce larger PDCs, you may need to model them separately

The “proximal hazard zone” on the map



Simplified volcano-hazards-zonation map for Mount Hood, Oregon. For more information on hazard zones see text or for a more detailed map and description see Scott and others, 1997, Volcano Hazards in the Mount Hood Region, Oregon, U.S. Geological Survey Open-File Report 97-89.

- In LAHARZ materials, and older USGS maps, it has been called the “proximal hazard zone”, but lots of recent user engagement work has shown that no one knows that that means!
- Maybe “maximum”, “multi-hazard”, or “near-volcano” hazard zone is easier to understand?

Work in progress: Update of the Mount Baker Long-Term Hazard Map

Volcano Hazard Zones

Hazards that can occur on or near the volcano

Name this Zone!!
Area where explosions, ash, lava, lahars, falling rocks, pyroclastic flows, and landslides can all occur within approximately 12 km of the summit vent

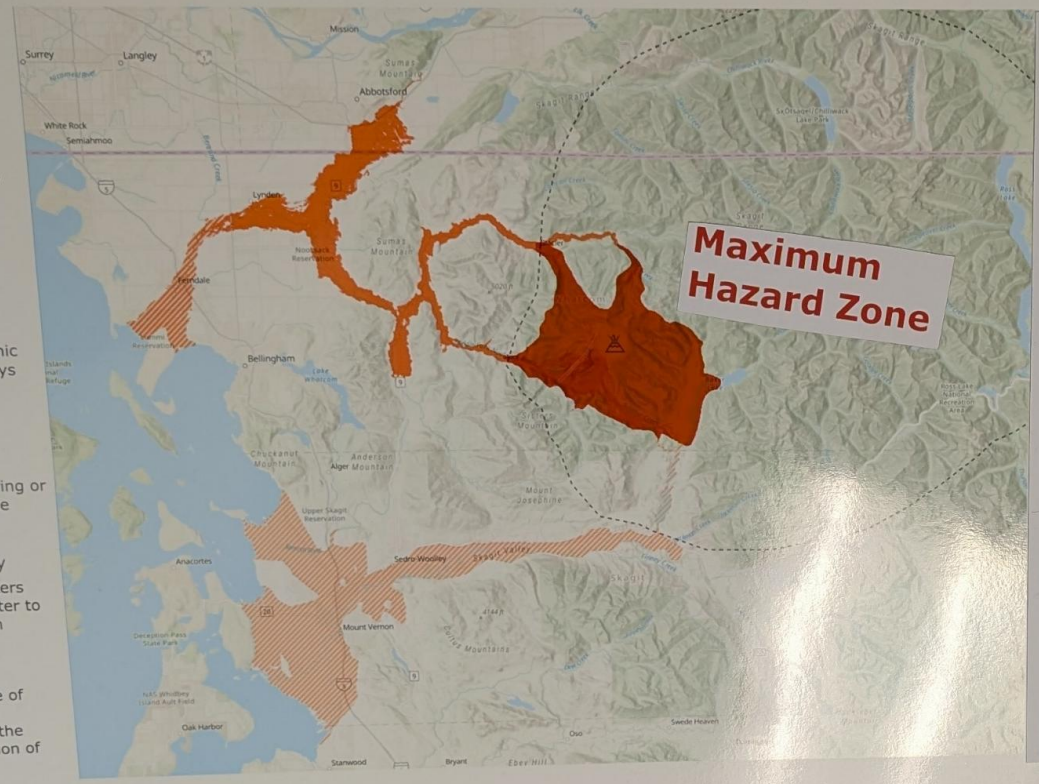
Hazards that can travel far from the volcano

Lahars
Area potentially inundated by voclanic mudflows traveling down river valleys originating on the volcano

Flooding in Nooksack River Lowlands
Area downriver that may flood during or after large-volume lahars reach the Nooksack River basin

Flooding in Skagit River Valley
Area that may flood if a lahar enters Baker Lake Reservoir causing water to overtop or erode around the dam

Ashfall
Area with at least a 25% chance of moderate (≥0.25 inches) ash accumulation from an eruption the size of the largest known eruption of Mount Baker



Volcano Hazards at Mount Baker

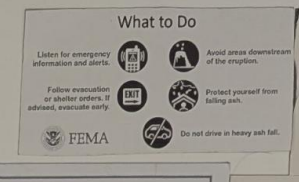
Lava Flows
Lava is molten rock (magma) that pours or oozes onto the Earth's surface. Numerous eruptions of lava interbedded with rock rubble constructed Mount Baker.

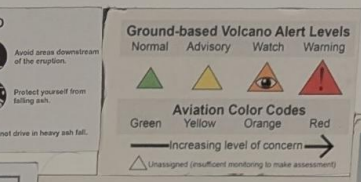
Pyroclastic Flows
Pyroclastic flows are hot avalanches of lava fragments and volcanic gas formed by the collapse of lava flows or eruption clouds.

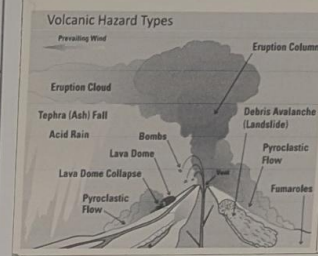
Tephra
Explosive eruptions blast fragments of rock high into the air. Large fragments fall to the ground close to the volcano. Small fragments (called ash) from the largest eruptions can travel hundreds of miles.

Lahars
Lahars are fast-moving slurries of rock, mud, and water that look and behave like flowing wet concrete.

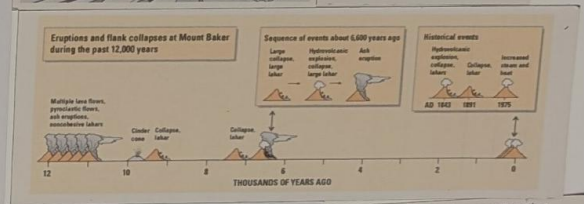
Landslides can transform into lahars. Pyroclastic flows can generate lahars by melting snow and ice.

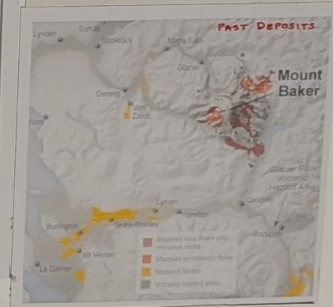




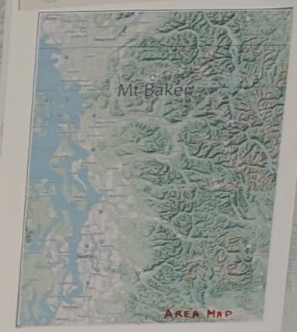










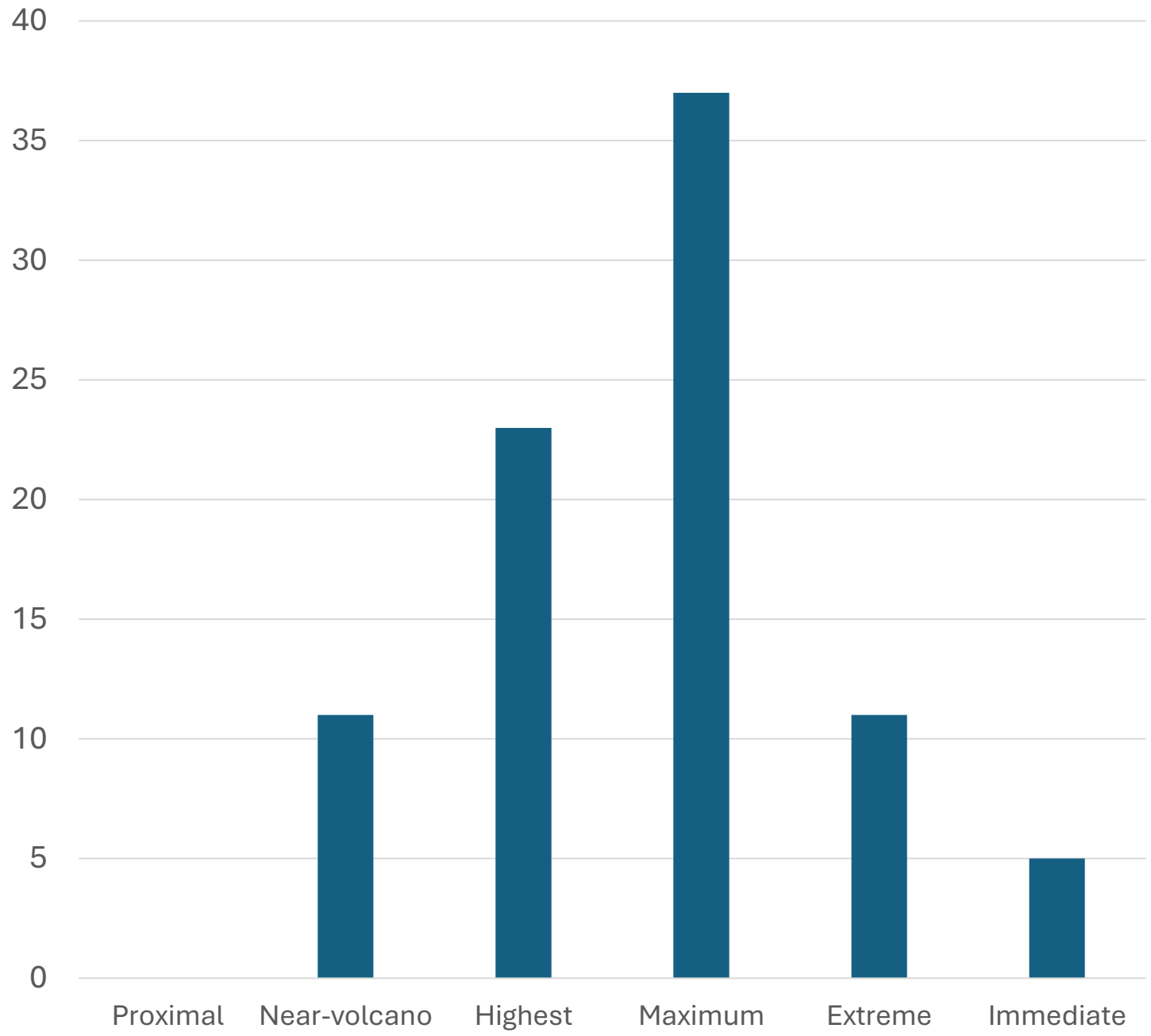


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For more information:
www.usgs.gov/volcanoes/mount-baker

85 people voted (mostly adults and older kids)

	Votes #/%
Proximal Hazard Zone	0/0%
Near-Volcano Hazard Zone	11/13%
Highest Hazard Zone	23/26%
Maximum Hazard Zone	37/43%
Extreme Hazard Zone	11/13%
Immediate Hazard Zone	5/6%



85 people voted (mostly adults and older kids)

Votes #/%

Proximal Hazard Zone

0/0%

About 5 people volunteered that they knew what it meant

Near-Volcano Hazard Zone

More than 5 people volunteered that they didn't know what it meant

Highest Hazard Zone

Everyone who commented on this name said "people don't know what it means"

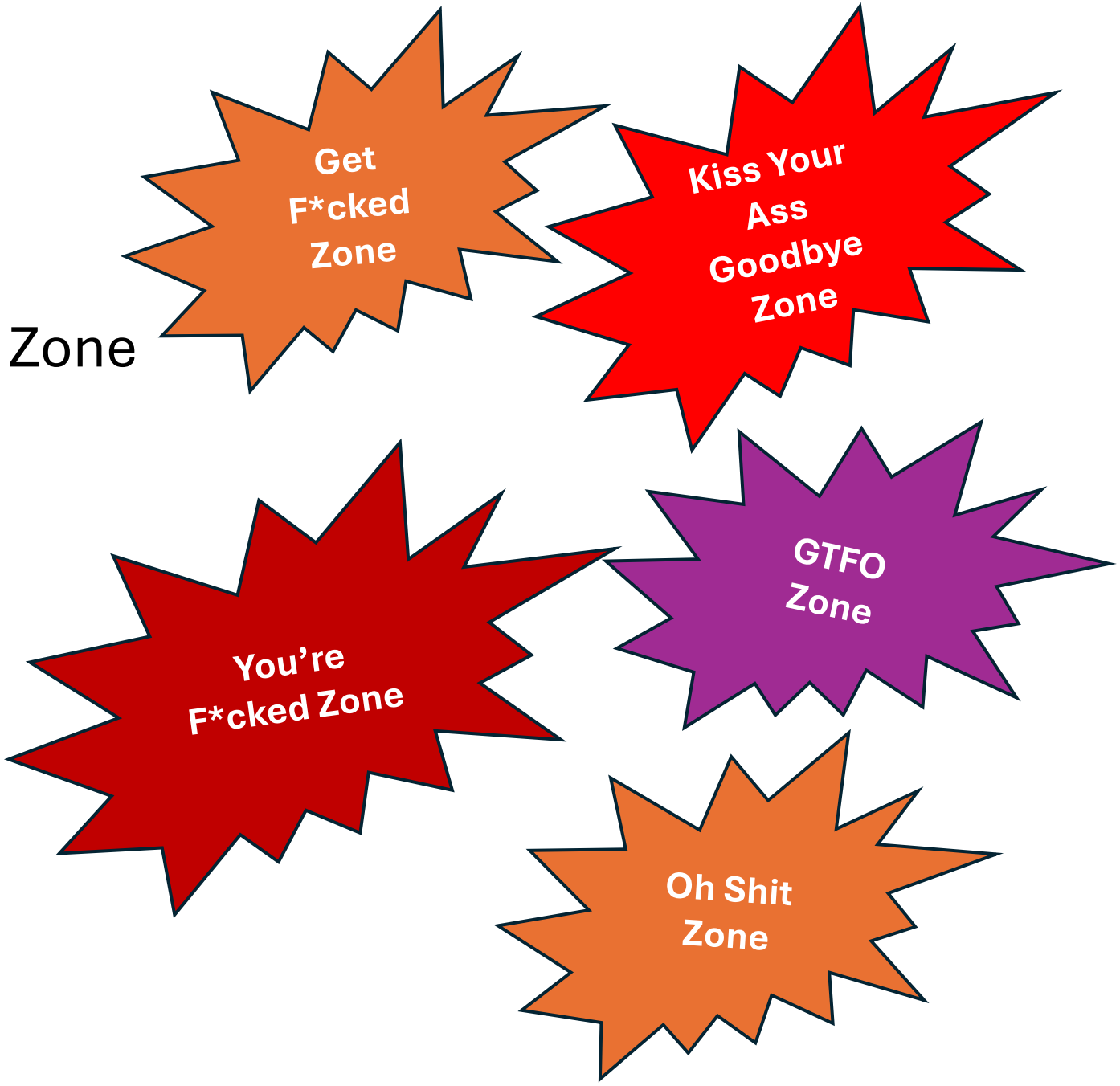
Maximum Hazard Zone

Had a few teachers and former journalists mention the guidance to design for an eighth-grade audience

Extreme Hazard Zone

Immediate Hazard Zone

- Danger Zone 🎶 🎸
- Get Out Zone
- No Joke, Get Out Zone
- Super Crazy Get Out of Here Zone
- No Survival Zone
- No-No Zone
- Multiple Lethal Hazard Zone
- NOPE Zone
- Very Hot Volcano Zone
- 💀



What is the message?

- Location: Near-volcano hazard zone
- Severity: Maximum or Extreme hazard zone
- Timing: Immediate Hazard Zone
- Number of Hazards/Severity: Multi-hazard Zone